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TwitterClone: Web Application

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DECLARATION

The Work reported in this project was carried by me under the supervision of **Mr. Muhammad Bilal** at **Govt Postgraduate Islamia College Sargodha Road Faisalabad**.

I hereby declare that this project and contents of the project are the product of my own work and is not done by any other person.

I further declare that this work has not been submitted for an award of any other degree. The institution may act if the information provided is found inaccurate at any stage.

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ABSTRACT

TwitterClone is a comprehensive social networking web application developed using PHP, MySQL, JavaScript, jQuery, Bootstrap, and AJAX. It allows users to connect with others, share tweets, upload media, follow accounts, send direct messages, and interact with content through likes, comments, replies, retweets, quote tweets, bookmarks, and notifications. Designed with a user-friendly approach, TwitterClone provides a familiar Twitter/X-like experience for online communication and content sharing.

The platform offers several advanced features, including user profile management, real-time-style direct messaging, media-based tweets, hashtag trends, explore feeds, and short-form reels with optional music support. Users can manage their accounts, update profiles, upload images and videos, save posts through bookmarks, and discover content through search and trending hashtags. The system also includes an admin dashboard for managing users, tweets, comments, replies, quotes, bookmarks, reels, music, trends, and platform analytics. By combining social interaction, media sharing, messaging, reels, and admin moderation, TwitterClone provides a complete and practical social media platform.

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CHAPTER NO 1 INTRODUCTION

1.1 Introduction to Project

TwitterClone is a browser-based social networking platform designed to provide users with a Twitter/X-like digital communication experience. The system allows users to create short posts, follow other users, interact through likes, comments, replies, retweets, and quote tweets, and communicate privately through direct messaging.

The project is developed using **PHP** as the backend technology and **MySQL** as the database system. It uses server-rendered pages, PHP sessions for authentication, PDO for database interaction, and JavaScript/jQuery for interactive features such as AJAX-based likes, follows, comments, bookmarks, search, messaging, and reels interactions.

Although the project started as a Twitter-style clone, it has expanded into a broader social media platform with advanced features such as direct messaging, bookmarks, notifications, admin moderation, short-form reels, music-backed reel creation, personalized discovery, and analytics.

The platform caters to general users, content creators, and administrators by providing tools for social interaction, content sharing, media publishing, private communication, and platform management.

1.2 Purpose of Project

The main purpose of TwitterClone is to provide a self-hosted social media platform where users can share thoughts, media, and short-form content while interacting with other users in a familiar social networking environment.

The project aims to solve the need for a lightweight Twitter-like platform that supports public posting, user engagement, private communication, content discovery, and administrative control. Users can create posts, follow accounts, send direct messages, explore trending hashtags, save posts through bookmarks, and consume or create reels.

For administrators, the system provides moderation and management features, allowing them to monitor users, tweets, comments, replies, quotes, bookmarks, trends, reels, music assets, and platform analytics.

The project also demonstrates practical implementation of real-world social media features using PHP, MySQL, JavaScript, AJAX, and supporting libraries.

1.3 How it Works

TwitterClone works as a session-based web application where users interact with the system through public pages, authenticated user pages, AJAX endpoints, and admin panels.

Users can perform the following actions:

- Register and log in using secure session-based authentication.
- Recover their password through OTP email verification using PHPMailer.
- Create and manage their profile, including name, username, bio, avatar, cover image, location, and website.
- Create tweets with text, images, and videos.
- Like, comment, reply, retweet, and quote tweets.
- Follow and unfollow other users.
- Search users and explore platform content.
- View trending hashtags and hashtag-based feeds.
- Save tweets using bookmarks.
- Receive notifications for likes, follows, retweets, comments, replies, quotes, and mentions.
- Send direct messages with text, files, documents, images, videos, and voice notes.
- Create and watch reels with optional music support.
- Interact with reels through likes, saves, comments, replies, follows, and views.
- Manage account settings such as username and password.

The application uses PHP files to render pages and process requests. Core business logic is handled through service classes such as User, Tweet, and Follow. Interactive actions are handled using AJAX endpoints located in folders such as `core/ajax`, `includes`, and some page-level controllers like `messages.php` and `reels.php`.

The admin panel works separately from the user area and allows platform administrators to manage users, tweets, replies, comments, quotes, trends, bookmarks, reels, music, and analytics.

1.4 Scope of Project

TwitterClone targets users who want to use a social networking platform for posting, communication, media sharing, and content discovery. The system is suitable for:

- General users who want to post, follow, like, comment, and message.
- Content creators who want to publish tweets, images, videos, and reels.
- Platform administrators who need tools to moderate users and content.
- Developers and students who want to understand how a social media platform works using PHP and MySQL.

The scope of the project includes:

- User registration, login, logout, and password recovery.
- Profile creation and profile customization.
- Tweet creation with text and media support.
- Social interaction through likes, comments, replies, retweets, quote tweets, and mentions.
- Follow and unfollow functionality.
- Notifications for user engagement.
- Search and explore functionality.
- Trending hashtags and topic discovery.
- Bookmarking system for saving tweets.
- Direct messaging with attachments, voice notes, replies, pins, deletion options, and blocking.
- Reels creation, reels feed, reels engagement, and music-backed reel publishing.
- Admin dashboard for moderation and analytics.
- Management of users, tweets, comments, replies, quotes, trends, bookmarks, reels, and music assets.

The project does not include e-commerce features such as products, carts, orders, checkout, inventory, or payment gateways.

1.5 Feasibility

TwitterClone is feasible as a web-based social networking application because it uses commonly available technologies, open-source tools, and a traditional PHP/MySQL architecture that can be deployed easily on local or shared hosting environments.

1.5.1 Technical Feasibility

The project is technically feasible because it is built using widely supported technologies such as PHP, MySQL/MariaDB, PDO, JavaScript, jQuery, Bootstrap, and Apache.

The backend is developed in PHP, while MySQL/MariaDB is used as the main database for storing users, posts, tweets, follows, likes, comments, replies, notifications, bookmarks, messages, reels, admin data, and personalization records.

The system uses PDO for database access, which provides prepared statements and a reusable database connection layer. JavaScript and jQuery are used to improve user interaction through AJAX-based operations such as likes, follows, comments, bookmarks, search, messaging, and reels actions.

The project also integrates useful libraries such as PHPMailer for OTP email delivery, Chart.js for admin analytics charts, Wavesurfer.js for audio waveform previews, Font Awesome for icons, Bootstrap for layout support, and FFmpeg for reel video audio processing.

1.5.2 Economic Feasibility

The project is economically feasible because it is developed using mostly open-source technologies. PHP, MySQL/MariaDB, Apache, Bootstrap, jQuery, Font Awesome, and many other supporting tools can be used without major licensing costs.

The application can run on a local XAMPP environment or a standard PHP-supported hosting service, which keeps development and deployment costs low.

The project can also support future monetization opportunities such as promoted content, premium profiles, creator tools, advanced analytics, or private community hosting, although these monetization features are not currently confirmed in the existing system.

1.5.3 Operational Feasibility

TwitterClone is operationally feasible because it provides a familiar social media interface similar to Twitter/X. Users can easily understand core actions such as posting tweets, following users, liking posts, commenting, messaging, and viewing notifications.

The system includes different modules for different user needs. Normal users can interact with social features, content creators can publish tweets and reels, and administrators can manage platform activity through the admin panel.

The use of server-rendered pages with AJAX interactions makes the application simpler to operate compared to a complex single-page application. However, some modules such as messaging and reels are large and may require careful maintenance in the future.

1.5.4 Information Feasibility

The project manages information related to users, posts, tweets, follows, likes, comments, replies, retweets, bookmarks, notifications, messages, reels, music assets, trends, and admin analytics.

The admin dashboard provides useful platform information such as activity charts, engagement breakdowns, top contributors, recent comments, trending hashtags, and user analytics.

The system also supports personalized content discovery through explore and reels recommendation logic, using information such as hashtags, categories, follows, engagement, and freshness.

User data is stored in a relational database, making the information structure understandable and manageable. However, the project analysis shows that some newer tables are not fully included in the original SQL dump, so schema normalization is recommended for better information management.

1.5.5 Motivational Feasibility

The project motivates users by providing multiple engagement features. Users are encouraged to post content, follow others, receive notifications, save posts, create reels, and interact through likes, comments, replies, retweets, and messages.

Content creators are motivated through richer media support, including multi-image tweets, multi-video tweets, reels, and music-backed reel creation. Users also receive feedback through notifications, likes, comments, and engagement actions.

The reels feed, explore page, trending hashtags, and recommendations increase user engagement and encourage users to spend more time on the platform.

1.5.6 Legal and Ethical Feasibility

The project can be legally and ethically feasible if proper privacy, security, and content moderation practices are followed.

The system includes admin moderation features for managing users, tweets, comments, replies, quotes, bookmarks, trends, reels, and music. It also supports blocking functionality in direct messages, which helps users control unwanted interactions.

However, the project should improve legal and ethical compliance by adding stronger privacy policies, clear terms of service, secure handling of user data, protection against unauthorized access, proper upload validation, CSRF protection, and audit logging for admin actions.

Hard-coded credentials, especially SMTP credentials, should be moved to environment variables to protect sensitive information.

1.6 Risk Management

Risk management strategies for TwitterClone include:

- **Data Security Risks:** Mitigated through secure session-based authentication, password hashing, prepared database queries, and proper protection of user accounts, messages, and media files.

- **Scalability Risks:** Addressed by improving database indexing, optimizing feed and reels queries, and separating large modules such as messaging and reels into more manageable components.
- **User Adoption Risks:** Managed by providing a familiar Twitter-like interface, simple navigation, responsive design, and interactive features such as likes, comments, notifications, bookmarks, and reels.
- **System Downtime Risks:** Minimized through regular database backups, proper server configuration, writable upload directories, updated database migrations, and separation of development and production settings.

1.7 Project Planning

The project follows an iterative development approach with continuous improvement of features. Key milestones include:

1. Requirements gathering and initial design of the social media platform.
2. Development of core features such as user authentication, profiles, tweets, likes, comments, and follows.
3. Integration of advanced features such as direct messaging, notifications, bookmarks, explore page, and reels.
4. Development of the admin dashboard for user management, content moderation, trends, reels, music, and analytics.
5. Testing and quality assurance of user flows, media uploads, messaging, reels, AJAX actions, and database operations.
6. Deployment, maintenance, and future updates based on performance, security, and user feedback.

1.8 Project Modules

The project consists of the following main modules:

1. **Authentication Module:** Handles user registration, login, logout, password reset, session management, and account security.

2. **Profile Management Module:** Enables users to view and update their profile information, including username, bio, profile image, cover image, location, and website.
3. **Tweet and Feed Module:** Allows users to create tweets, upload images or videos, view home feeds, profile feeds, status pages, and hashtag-based content.
4. **Engagement Module:** Manages likes, comments, replies, retweets, quote tweets, mentions, bookmarks, and notifications.
5. **Messaging Module:** Provides direct communication between users through text messages, attachments, voice notes, replies, pins, and blocking features.
6. **Reels Module:** Enables users to create, view, like, save, comment on, and interact with short-form reels with optional music support.
7. **Media Upload Module:** Handles profile images, cover images, tweet media, message attachments, reel media, and music assets.
8. **Admin Dashboard Module:** Provides tools for managing users, tweets, comments, replies, quotes, trends, bookmarks, reels, music, and platform analy

CHAPTER NO 2 BACKGROUND AND PROBLEM DEFINITION

2.1 Background Research

Social media platforms have become an important part of modern digital communication. Users now require online platforms where they can share thoughts, post media, follow others, communicate privately, and discover trending content. Applications like Twitter/X have created a strong demand for short-post-based social networking systems.

Existing platforms provide many social interaction features, but they are usually closed-source and cannot be customized easily for personal, educational, or small community use. This creates the need for a self-hosted social media application like TwitterClone, which provides core social networking features such as tweeting, following, messaging, notifications, bookmarks, reels, and admin moderation.

2.2 Existing Technology

Several technologies contribute to the development of social media platforms, including:

- **Social Networking Platforms:** Applications like Twitter/X, Facebook, and Instagram provide posting, following, messaging, and engagement features, but they are not fully customizable by users or developers.
- **Content Sharing Systems:** Platforms such as blogs and forums allow users to publish content, but they usually lack real-time interaction, reels, notifications, and personalized feeds.
- **Messaging Applications:** Apps like WhatsApp and Messenger provide private communication, but they do not offer public timelines, hashtags, trends, or social feeds.

TwitterClone uses PHP, MySQL, JavaScript, jQuery, Bootstrap, PHPMailer, Chart.js, and FFmpeg to provide a complete social media experience with posting, interaction, messaging, reels, media uploads, admin control, and analytics.

2.3 Area of Study

The primary area of study for TwitterClone involves:

- **Web Development:** Building a dynamic social media application using PHP, MySQL, HTML, CSS, JavaScript, and AJAX.
- **Database Management:** Storing and managing users, tweets, likes, comments, replies, follows, messages, notifications, reels, and admin data.
- **User Interface Design:** Creating a familiar Twitter-like interface with feeds, sidebars, profiles, search, notifications, messages, and reels.
- **Social Media Interaction:** Implementing features such as posting, following, liking, commenting, replying, retweeting, bookmarking, and messaging.
- **Admin and Analytics:** Providing an admin dashboard for managing users, content, reels, music, trends, and platform activity.

2.4 Reason for the Project

The reason for developing TwitterClone is to create a customizable and self-hosted social media platform that demonstrates the working of a modern social networking system. Many existing platforms do not allow developers to fully control features, database structure, moderation tools, or user experience.

TwitterClone fills this gap by providing essential social media features in one application, including tweets, profiles, followers, notifications, bookmarks, messaging, reels, trends, and admin management. It also helps in understanding how real-world social platforms manage users, content, interactions, media, and moderation.

2.5 Objectives of the Project

The objectives of TwitterClone are:

1. **To Provide User Authentication:** Allowing users to register, log in, log out, and recover passwords securely.
2. **To Enable Social Posting:** Allowing users to create tweets with text, images, and videos.
3. **To Support User Interaction:** Providing likes, comments, replies, retweets, quote tweets, follows, mentions, bookmarks, and notifications.
4. **To Provide Direct Messaging:** Enabling users to communicate privately through messages, attachments, and voice notes.

5. **To Add Reels Functionality:** Allowing users to create, view, like, save, and comment on short-form reels with optional music support.
6. **To Provide Admin Control:** Giving administrators tools to manage users, tweets, comments, replies, quotes, trends, reels, music, bookmarks, and analytics.
7. **To Improve Content Discovery:** Supporting search, explore feeds, trending hashtags, personalized reels, and user suggestions.

CHAPTER NO 3 SYSTEM REQUIREMENT ANALYSIS

3.1 System Fundamental Requirements

The fundamental requirements of TwitterClone focus on providing a complete social media experience with secure user access and interactive features. These include:

- **User Authentication:** Secure user registration, login, logout, password reset, and session management.
- **Profile Management:** Allow users to create, view, and update their profile details such as username, bio, profile image, cover image, location, and website.
- **Tweet Management:** Enable users to create tweets with text, images, and videos, and display them on home feeds, profiles, status pages, and hashtag pages.
- **Social Interaction:** Provide features such as likes, comments, replies, retweets, quote tweets, mentions, follows, bookmarks, and notifications.
- **Direct Messaging:** Allow users to send private messages, attachments, voice notes, replies, and manage conversations.
- **Reels Management:** Enable users to create, view, like, save, comment on, and interact with short-form reels with optional music support.
- **Search and Discovery:** Provide user search, explore feeds, trending hashtags, user suggestions, and personalized content discovery.
- **Admin Dashboard:** Allow administrators to manage users, tweets, comments, replies, quotes, bookmarks, trends, reels, music, and platform analytics.
- **Responsive Design:** Ensure the system works properly on desktops, laptops, tablets, and mobile devices.

3.2 System Non-Fundamental Requirements

The non-fundamental requirements focus on improving the performance, security, reliability, and usability of TwitterClone.

3.2.1 Performance

- Ensure fast page loading and smooth user interactions.
- Optimize database queries for feeds, reels, search, and messaging.
- Use AJAX for interactive actions such as likes, follows, comments, bookmarks, and messaging.

3.2.2 3.2.2 Safety

- Validate user input to prevent invalid or harmful data.
- Protect uploaded media files through proper file type and size validation.
- Prevent unauthorized access to private messages and admin pages.

3.2.3 Security

- Use secure password hashing for user and admin accounts.
- Protect user sessions from unauthorized access.
- Use prepared statements to prevent SQL injection.
- Store sensitive credentials securely outside the source code.

3.2.4 Flexibility

- Allow future addition of new features such as improved recommendations, content moderation, and advanced analytics.
- Support separate roles for normal users and administrators.

3.2.5 Portability

- Ensure the application can run on common PHP-supported environments such as XAMPP or Apache servers.
- Support modern browsers such as Chrome, Firefox, Edge, and Safari.

3.2.6 Scalability

- Design the database and modules to handle increasing users, tweets, messages, notifications, and reels.
- Add indexing, caching, and optimized queries for better performance.

3.2.7 Reliability

- Maintain stable system performance during normal usage.
- Use backups to protect user data and platform content.
- Reduce errors caused by missing tables or configuration issues.

3.2.8 Usability

- Provide a familiar Twitter-like interface with clear navigation.
- Make features such as posting, messaging, searching, and viewing notifications easy to use.
- Ensure responsive layouts for different screen sizes.

3.2.9 Maintainability

- Keep code organized into reusable classes, handlers, includes, and AJAX files.
- Improve large modules such as messaging and reels by separating logic into smaller components.
- Use version control for tracking changes.

3.2.10 Reusability

- Reuse common UI components such as tweet cards, search boxes, sidebars, profile dropdowns, and trend sections.
- Reuse service classes for user, tweet, and follow-related operations.

3.2.11 Efficiency

- Optimize database operations to reduce server load.
- Use lightweight frontend libraries such as jQuery and Bootstrap.
- Avoid unnecessary repeated queries in feeds and sidebars.

3.3 Hardware and Software Requirements

3.3.1 Hardware Requirements

Client-Side:

- **Device:** Desktop, laptop, tablet, or smartphone.
- **Processor:** Minimum dual-core processor.
- **Memory:** At least 2 GB RAM.
- **Storage:** 200 MB free space for browser cache and temporary files.
- **Network:** Stable internet connection.

Server-Side:

- **Processor:** Quad-core processor or better.
- **Memory:** Minimum 4 GB RAM recommended.
- **Storage:** 50 GB SSD or higher for database, uploaded media, reels, and backups.
- **Network:** Stable high-speed internet connection for smooth access and media handling.

3.3.2 Software Requirements

Client-Side:

- **Operating System:** Windows, macOS, Linux, Android, or iOS.
- **Browser:** Latest version of Chrome, Firefox, Safari, or Microsoft Edge.

Server-Side:

- **Operating System:** Windows or Linux-based server.
- **Web Server:** Apache server with `.htaccess` and rewrite support.
- **Backend Language:** PHP.
- **Database:** MySQL or MariaDB.
- **Database Access:** PDO.
- **Package Manager:** Composer.
- **Libraries and Tools:** PHPMailer, jQuery, Bootstrap, Font Awesome, Chart.js, Wavesurfer.js, emoji-picker-element, and FFmpeg.
- **Local Development Environment:** XAMPP or any PHP/MySQL-supported server environment.

CHAPTER NO 4 SYSTEM DESIGN

4.1 Use Case Diagram

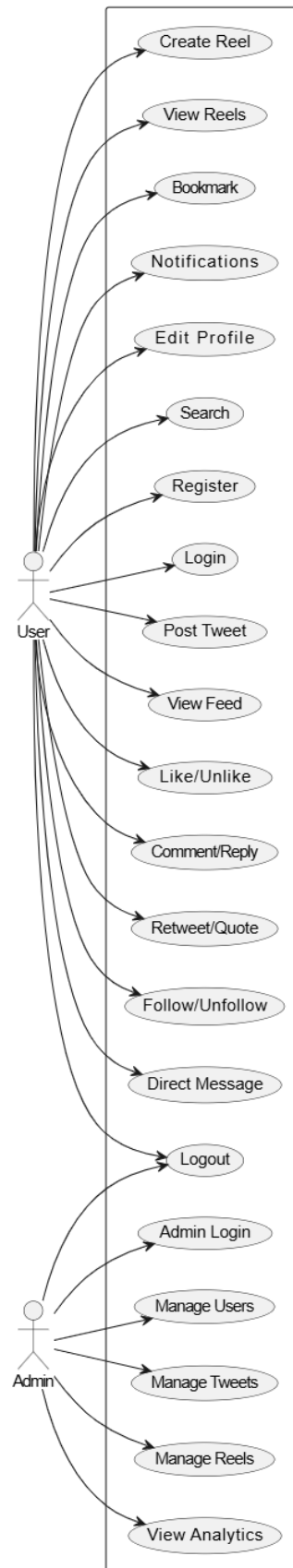


Figure 4-1 Use Case Diagram

4.2 Use Cases

4.2.1 Signup

Table 4.1 Signup Use Case

Identifier	UC-1	
Purpose	Registration	
Priority	High	
Pre-condition	The user is not already registered, and the system must be connected to the server/database.	
Post-condition	User is registered successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User clicks on the Signup button.	System displays the Signup form.
2	User enters name, username, email, password, and confirm password.	System processes the input for validation.
3	User clicks on the Signup button.	System saves user data and confirms success.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user password and confirm password are not the same.	System does not register the user and shows an error.
2	If any required field is left empty.	System does not register the user and shows an error.

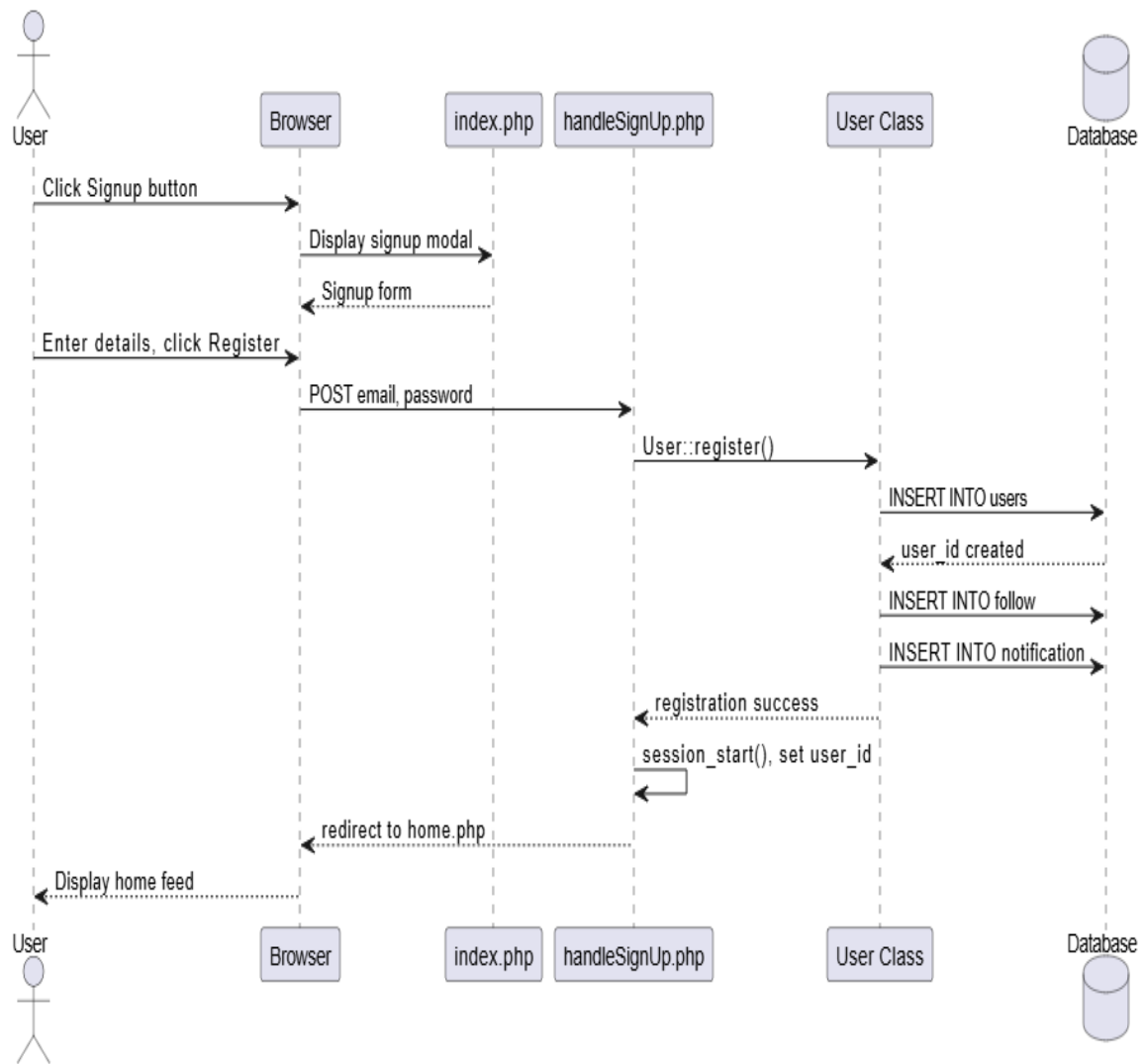


Figure 4-2Sequence Diagram of Signup

4.2.2 Login

Table 4.2 Login Use Case

Identifier	UC-2	
Purpose	Login	
Priority	High	
Pre-condition	The user must be registered, and the system must be connected to the server/database.	
Post-condition	User is logged in successfully and redirected to the home feed.	
Typical Course of Action		
S#	Actor Action	System Response
1	User clicks on the Login button.	System displays the Login form.
2	User enters email and password.	System processes the input for validation.
3	User clicks on the Login button.	System verifies the credentials from the database and grants access to the system.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the email or password is incorrect.	System does not log in the user and shows an error.
2	If any required field is left empty.	System does not log in the user and shows an error.

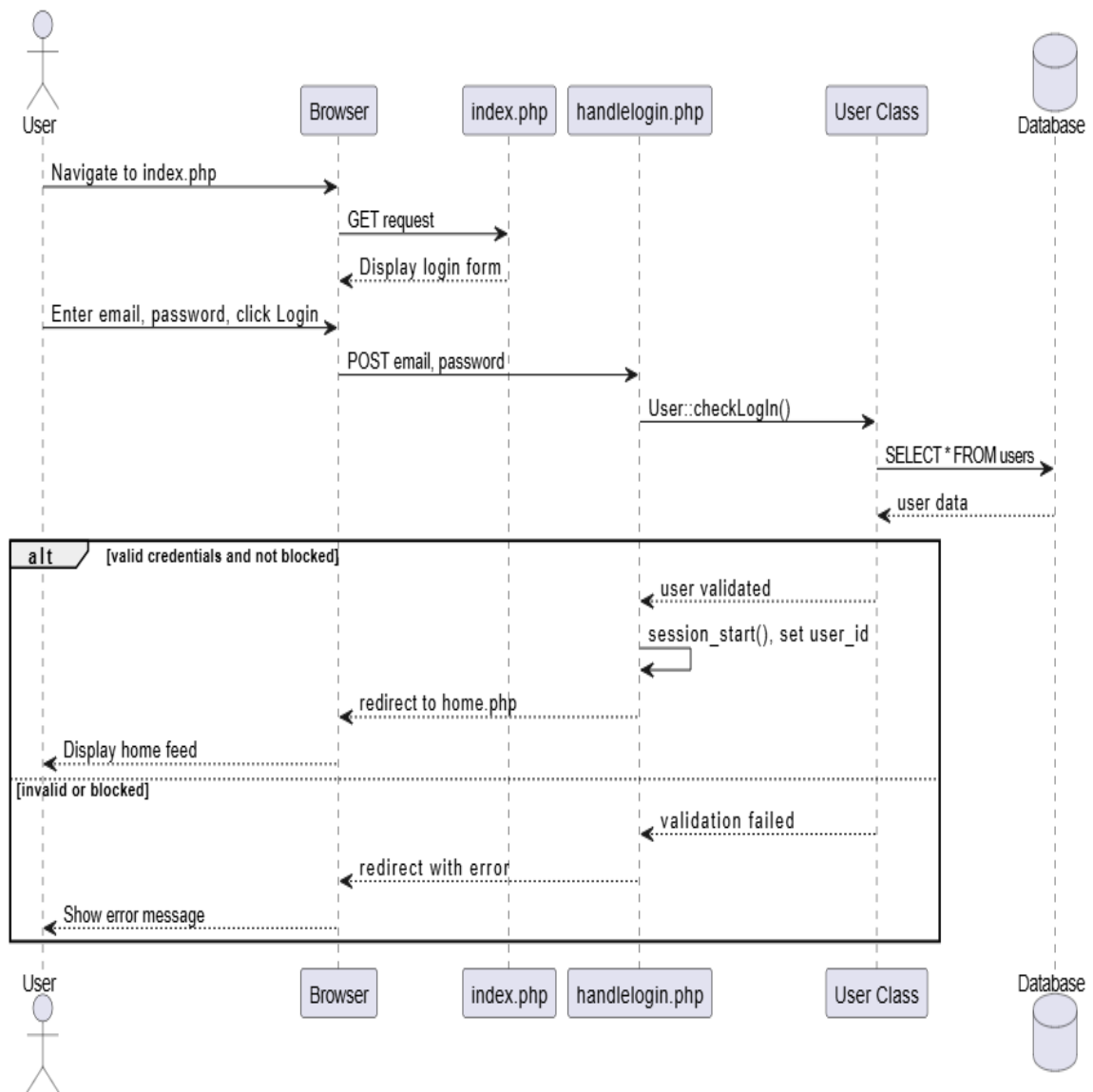


Figure 4-3 Sequence Diagram of Login

4.2.3 Create Tweet

Table 4.3 Create tweet Use Case

Identifier	UC-3	
Purpose	Create Tweet	
Priority	High	
Pre-condition	The user must be logged in, and the system must be connected to the server/database.	
Post-condition	Tweet is created successfully and displayed on the home feed/profile feed.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to the Home page.	System displays the tweet composer.
2	User enters tweet text and optionally selects image or video media.	System processes the input and validates the tweet content.
3	User clicks on the Tweet/Post button.	System saves the tweet data and media in the database.
4	Tweet is created successfully.	System displays the new tweet on the feed.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user leaves tweet content empty and does not upload media.	System does not create the tweet and shows an error.
2	If the uploaded media file type is invalid.	System rejects the file and shows an error.
3	If the tweet text exceeds the allowed limit.	System does not create the tweet and shows a validation error.

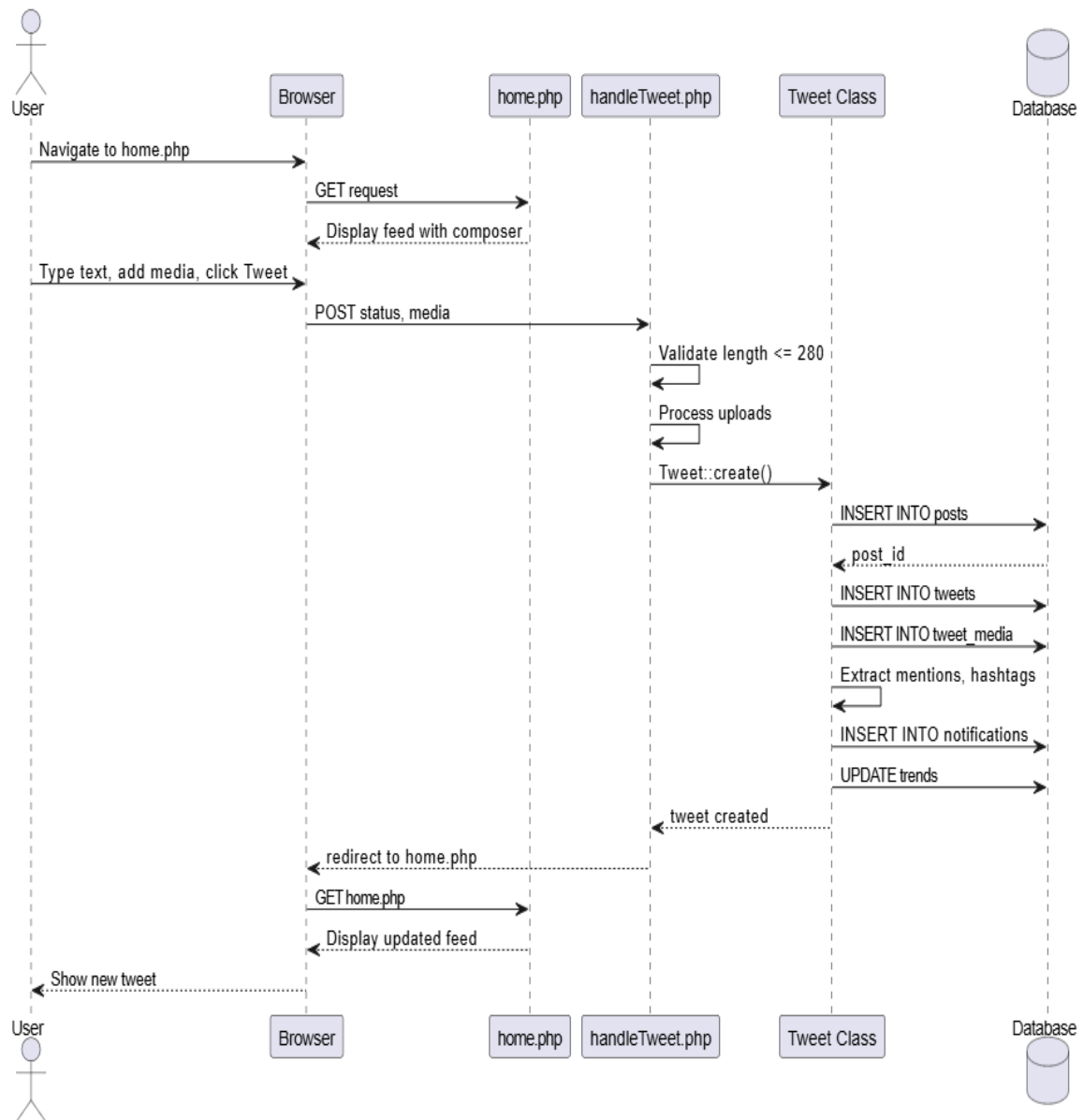


Figure 4-4 Sequence Diagram of Creating Tweet

4.2.4 View and Update Profile

Table 4.4 View and Update Profile Use Case

Identifier	UC-4	
Purpose	View and Update Profile	
Priority	High	
Pre-condition	The user must be logged in, and the system must be connected to the server/database.	
Post-condition	User profile is viewed or updated successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to the Profile page.	System displays the user profile details.
2	User clicks on the Edit Profile button.	System displays the profile update form.
3	User updates profile details such as name, bio, location, website, profile image, or cover image.	System processes the input and validates the updated data.
4	User clicks on the Save/Update button.	System saves the updated profile data in the database.
5	Profile is updated successfully.	System displays the updated profile information.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user leaves required fields empty.	System does not update the profile and shows an error.
2	If the uploaded image file type is invalid.	System rejects the file and shows an error.
3	If the entered website URL is invalid.	System does not save the URL and shows a validation error.

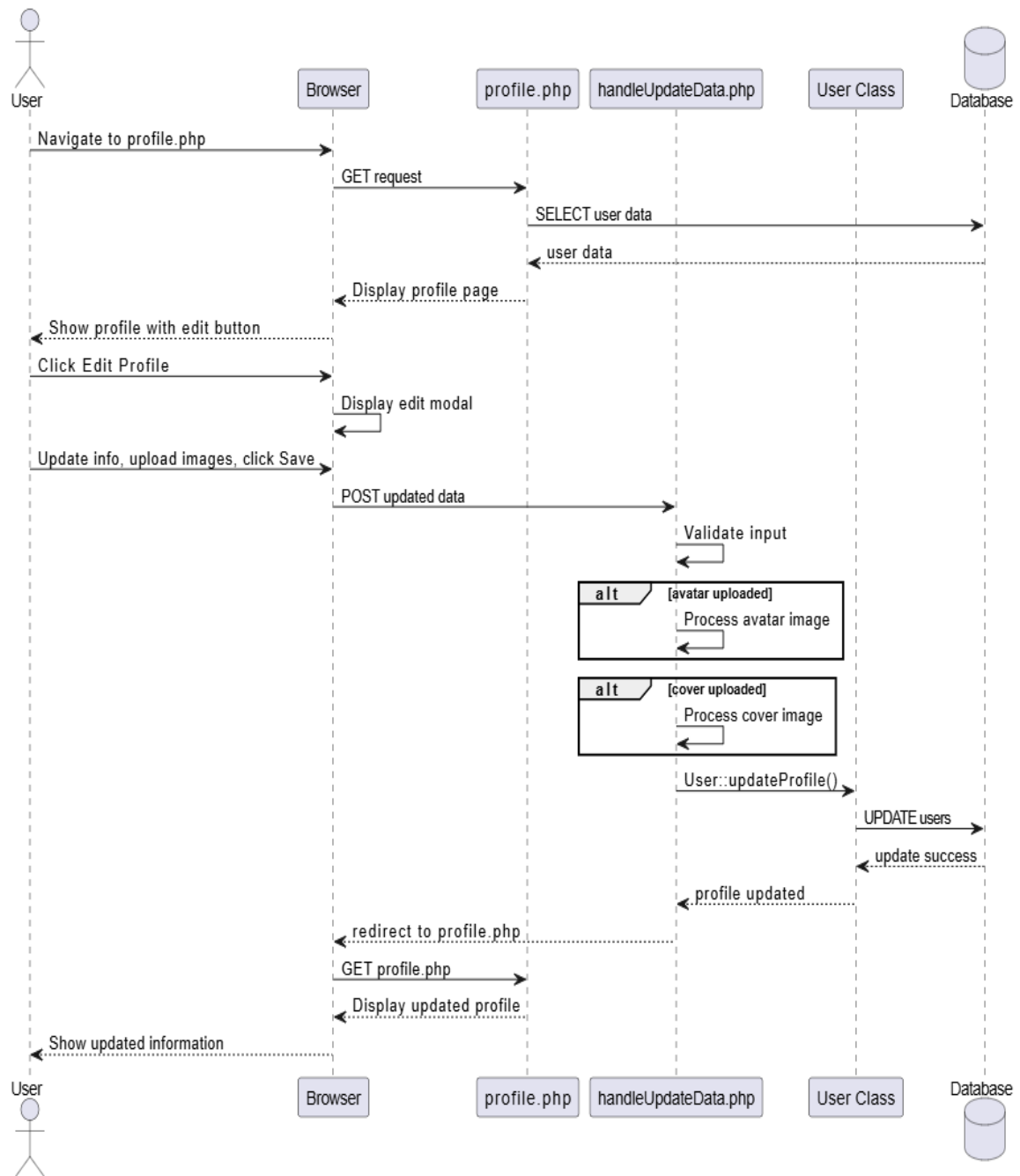


Figure 4-5 Sequence Diagram of View and Update Profile

4.2.5 Follow and Unfollow User

Table 4.5 Follow and Unfollow User Use Case

Identifier	UC-5	
Purpose	Follow and Unfollow User	
Priority	High	
Pre-condition	The user must be logged in, and the target user profile must exist in the system.	
Post-condition	Follow or unfollow action is completed successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to another user’s profile or suggestion section.	System displays the user profile or suggested user list.
2	User clicks on the Follow button.	System creates a follow relationship between both users.
3	User clicks on the Unfollow button.	System removes the follow relationship from the database.
4	Follow or unfollow action is completed.	System updates follower and following counts.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user tries to follow an invalid or deleted account.	System does not perform the action and shows an error.
2	If the user is already following the account.	System displays the Unfollow option instead of duplicating the follow action.
3	If the database connection fails.	System does not update the follow status and shows an error.

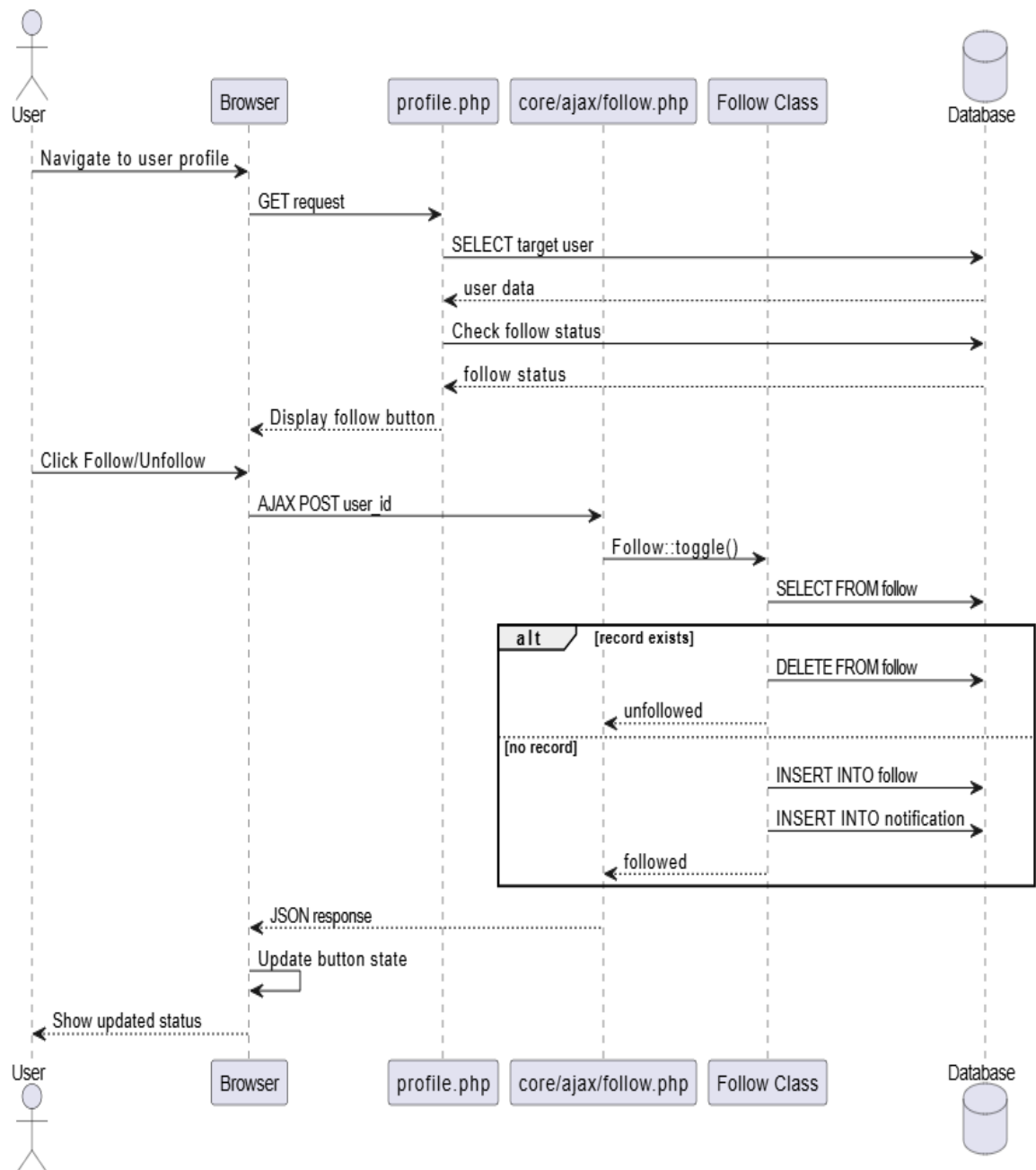


Figure 4-6 Sequence Diagram of Follow and Unfollow User

4.2.6 Like and Comment on Tweet

Table 4.6 Like comment on Tweet Use Case

Identifier	UC-6	
Purpose	Like Tweet	
Priority	Medium	
Pre-condition	The user must be logged in, and the tweet must exist in the system.	
Post-condition	Tweet like status and like count are updated successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User views a tweet on the home feed, profile feed, or status page.	System displays the tweet with the like button.
2	User clicks on the Like button.	System saves the like action in the database.
3	Like action is completed.	System updates the like count and changes the like button state.
4	Tweet owner receives interaction.	System generates a notification for the tweet owner.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user clicks the Like button again.	System removes the like and updates the like count.
2	If the tweet no longer exists.	System does not perform the action and shows an error.
3	If the database connection fails.	System does not update the like status and shows an error.

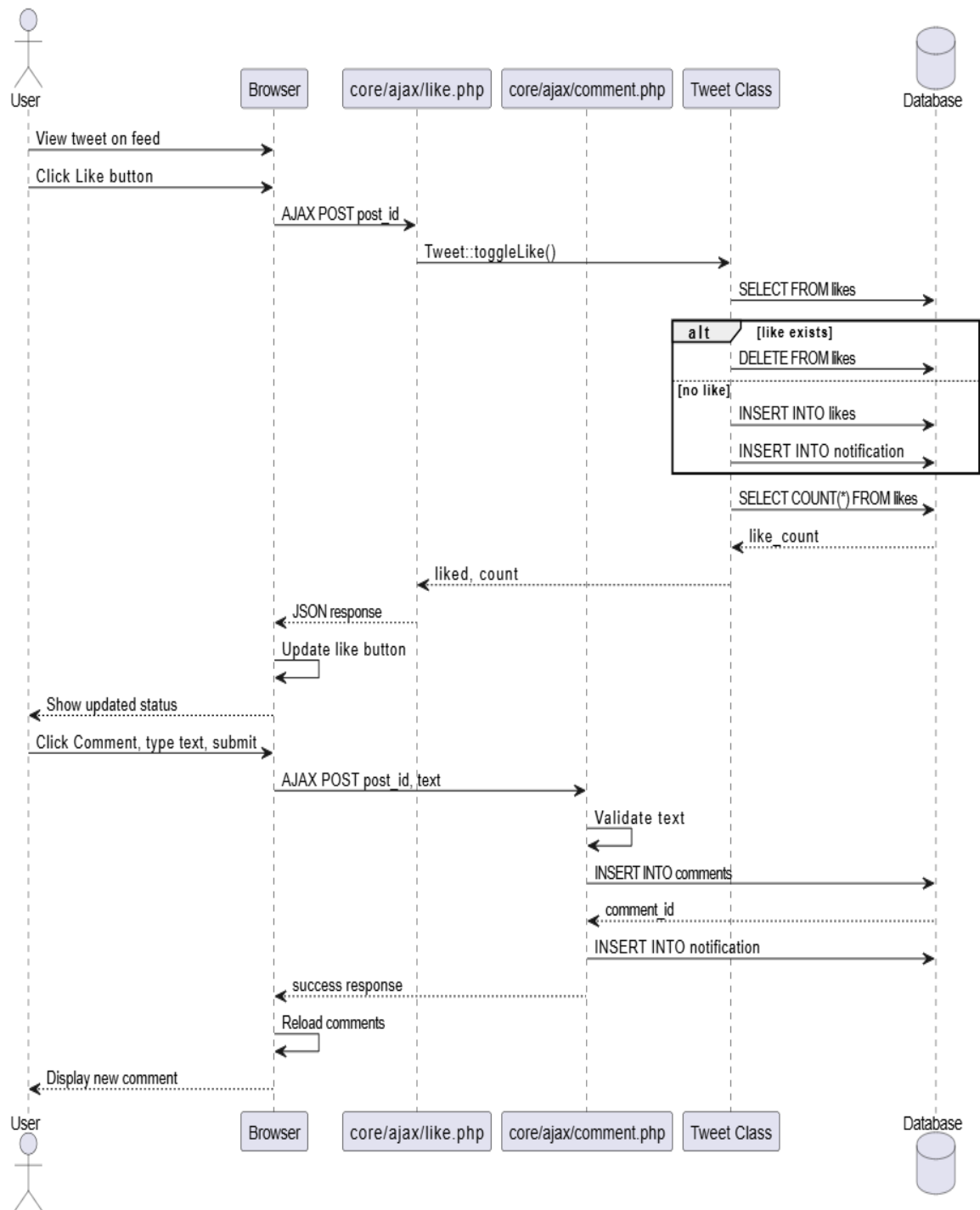


Figure 4-7 Sequence Diagram of Like and comment on Tweet

4.2.7 Direct Messaging

Table 4.7 Direct Messaging Use Case

Identifier	UC-7	
Purpose	Direct Messaging	
Priority	High	
Pre-condition	The user must be logged in, and the receiver must exist in the system.	
Post-condition	Message is sent successfully and displayed in the conversation.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to the Messages section.	System displays the message interface and conversation list.
2	User selects a receiver or existing conversation.	System loads the previous chat messages.
3	User enters a text message or selects an attachment/voice note.	System processes the message input for validation.
4	User clicks on the Send button.	System saves the message in the database and displays it in the chat.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the message field is empty and no attachment is selected.	System does not send the message and shows an error.
2	If the receiver does not exist.	System does not send the message and shows an error.
3	If either user has blocked the other user.	System prevents the message from being sent and shows a restriction message.

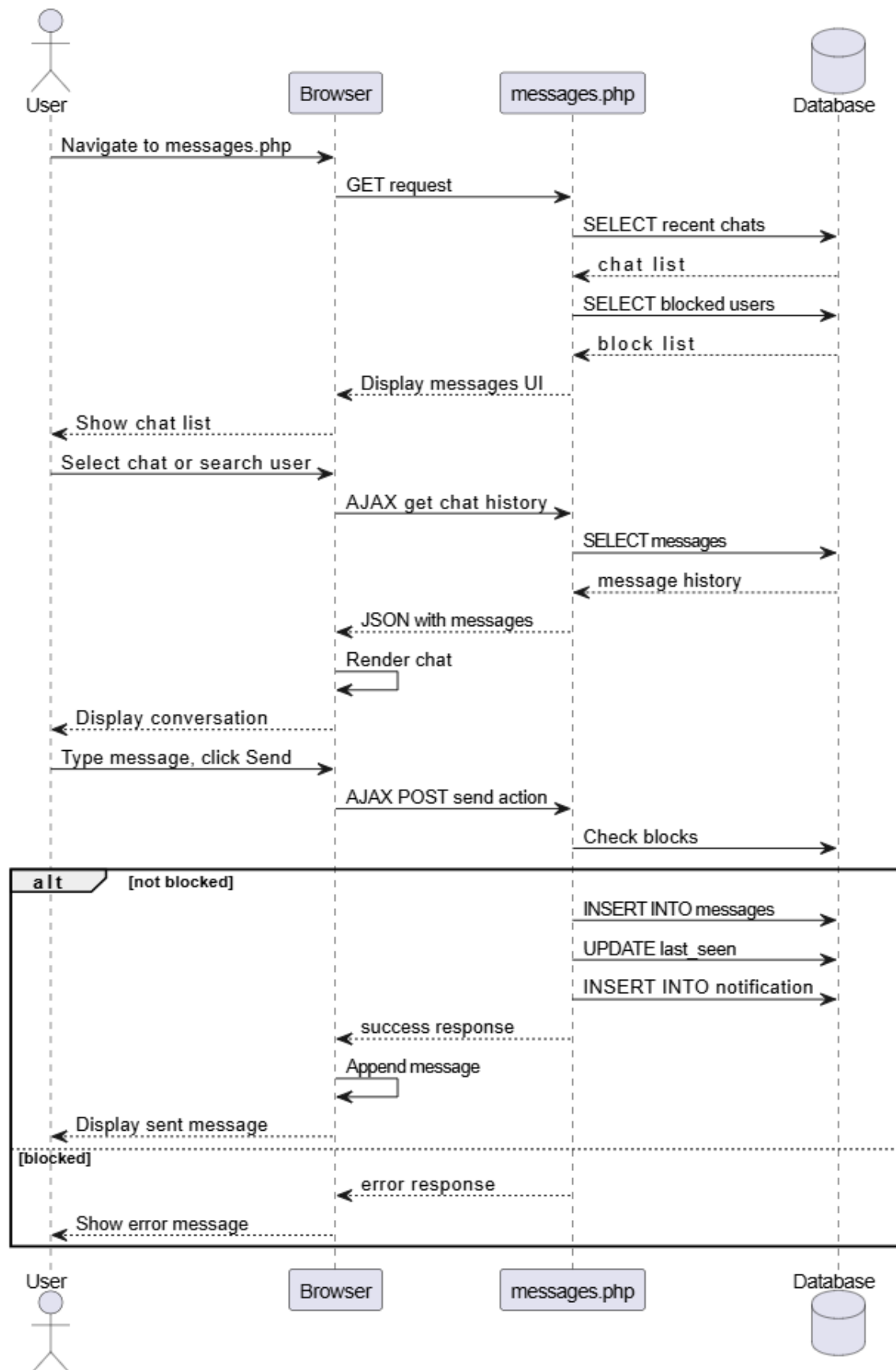


Figure 4-8 Sequence Diagram of Direct Messaging

4.2.8 Creating Reel

Table 4.8 Creating Reel Use Case

Identifier	UC-8	
Purpose	Create Reel	
Priority	High	
Pre-condition	The user must be logged in, and the system must be connected to the server/database.	
Post-condition	Reel is created successfully and displayed in the reels feed.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to the Create Reel section.	System displays the reel creation form.
2	User uploads an image or video for the reel.	System processes and validates the uploaded media file.
3	User adds caption and optionally selects background music.	System processes the reel details for validation.
4	User clicks on the Create/Post Reel button.	System saves the reel data in the database and uploads the media file.
5	Reel is created successfully.	System displays the reel in the reels feed.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the user does not upload any media file.	System does not create the reel and shows an error.
2	If the uploaded media file type is invalid.	System rejects the file and shows an error.
3	If the video reel exceeds the allowed duration.	System does not create the reel and shows a validation error.

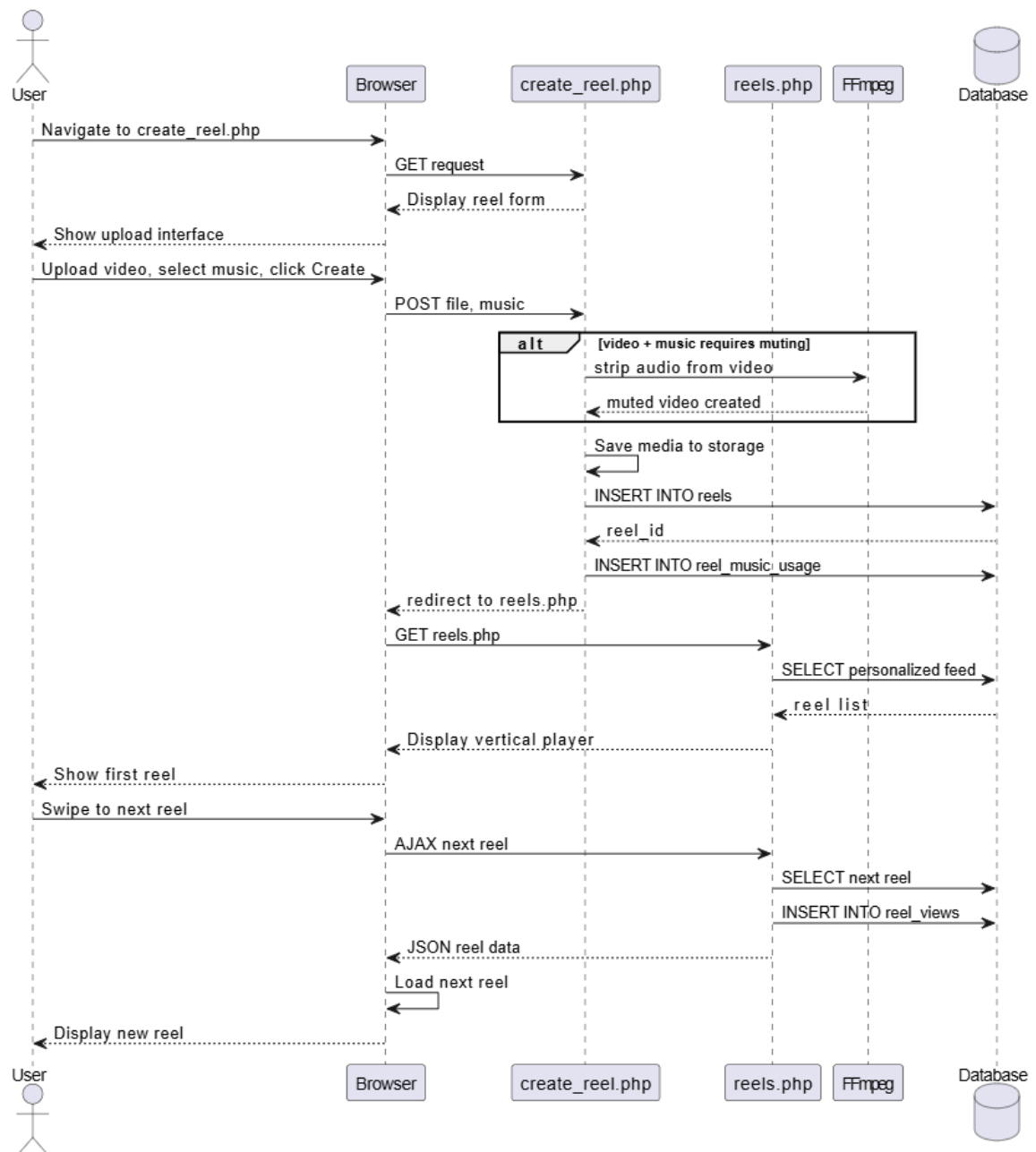


Figure 4-9 Sequence Diagram of Creating Reel

4.2.9 Check and Delete Notification

Table 4.9 Check and Delete Notification Use Case

Identifier	UC-9	
Purpose	Check and Delete Notification	
Priority	Medium	
Pre-condition	The user must be logged in, and notifications must exist in the system.	
Post-condition	Notifications are viewed, marked as read, or deleted successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User navigates to the Notifications section.	System displays all notifications related to the user.
2	User checks recent notifications.	System marks the notifications as viewed or read.
3	User clicks on the delete option for a notification.	System removes the selected notification from the database.
4	User clicks on delete all notifications.	System removes all notifications related to the user.
Alternate Course of Action		
S#	Actor Action	System Response
1	If no notifications are available.	System displays a no notifications message.
2	If the selected notification does not exist.	System does not delete it and shows an error.
3	If the database connection fails.	System does not update or delete notifications and shows an error.

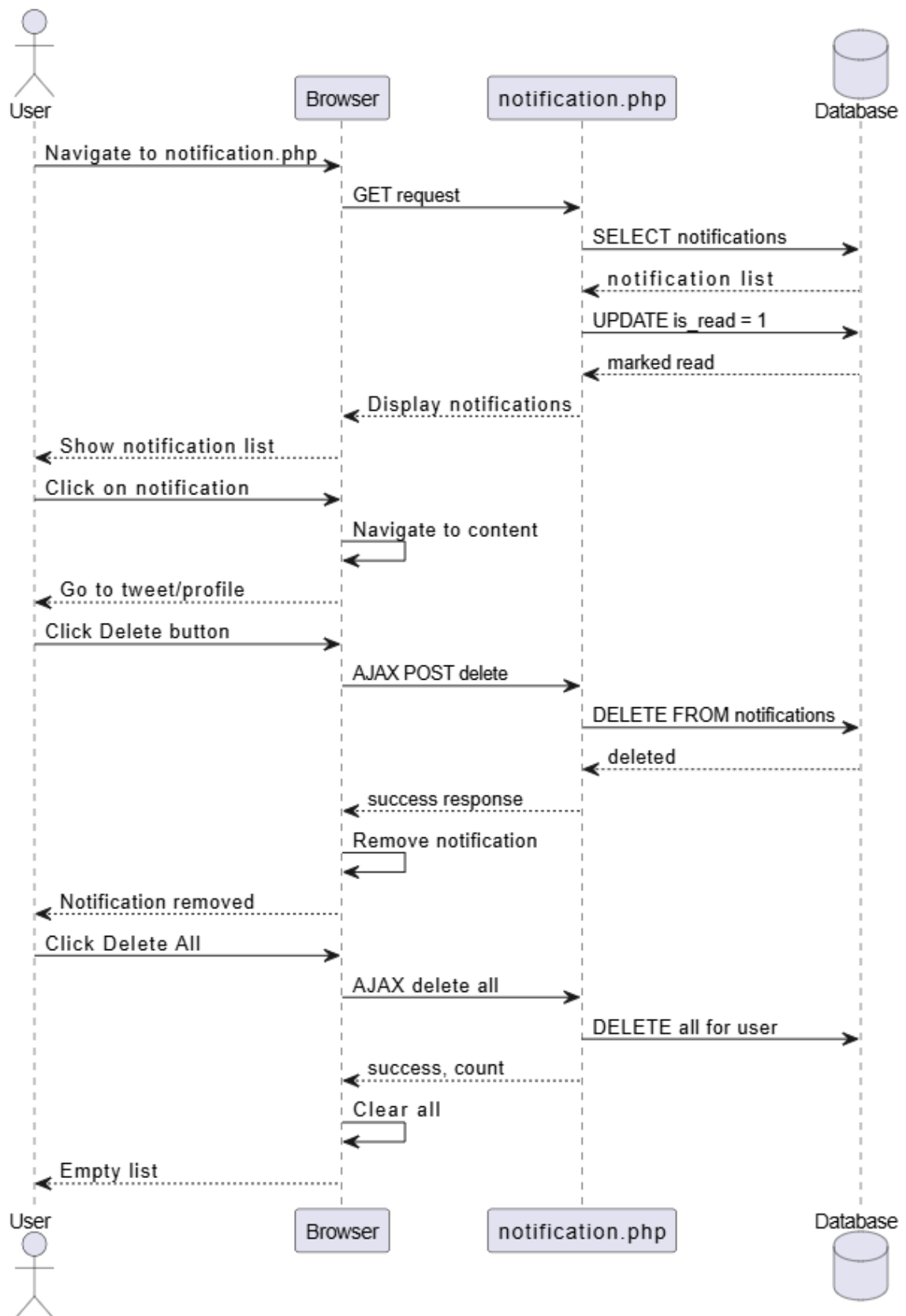


Figure 4-10 Sequence Diagram of Check and Delete Notification

4.2.10 Admin Login

Table 4.10 Admin Login Use Case

Identifier	UC-10	
Purpose	Admin Login	
Priority	High	
Pre-condition	Admin must be registered in the system, and the system must be connected to the server/database.	
Post-condition	Admin is logged in successfully and redirected to the admin dashboard.	
Typical Course of Action		
S#	Actor Action	System Response
1	Admin navigates to the Admin Login page.	System displays the admin login form.
2	Admin enters username/email and password.	System processes the input for validation.
3	Admin clicks on the Login button.	System verifies admin credentials from the database.
4	Admin credentials are valid.	System grants access and redirects admin to the dashboard.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the username/email or password is incorrect.	System does not log in the admin and shows an error.
2	If any required field is left empty.	System does not log in the admin and shows a validation error.
3	If a normal user tries to access the admin dashboard.	System denies access and redirects to the admin login page.

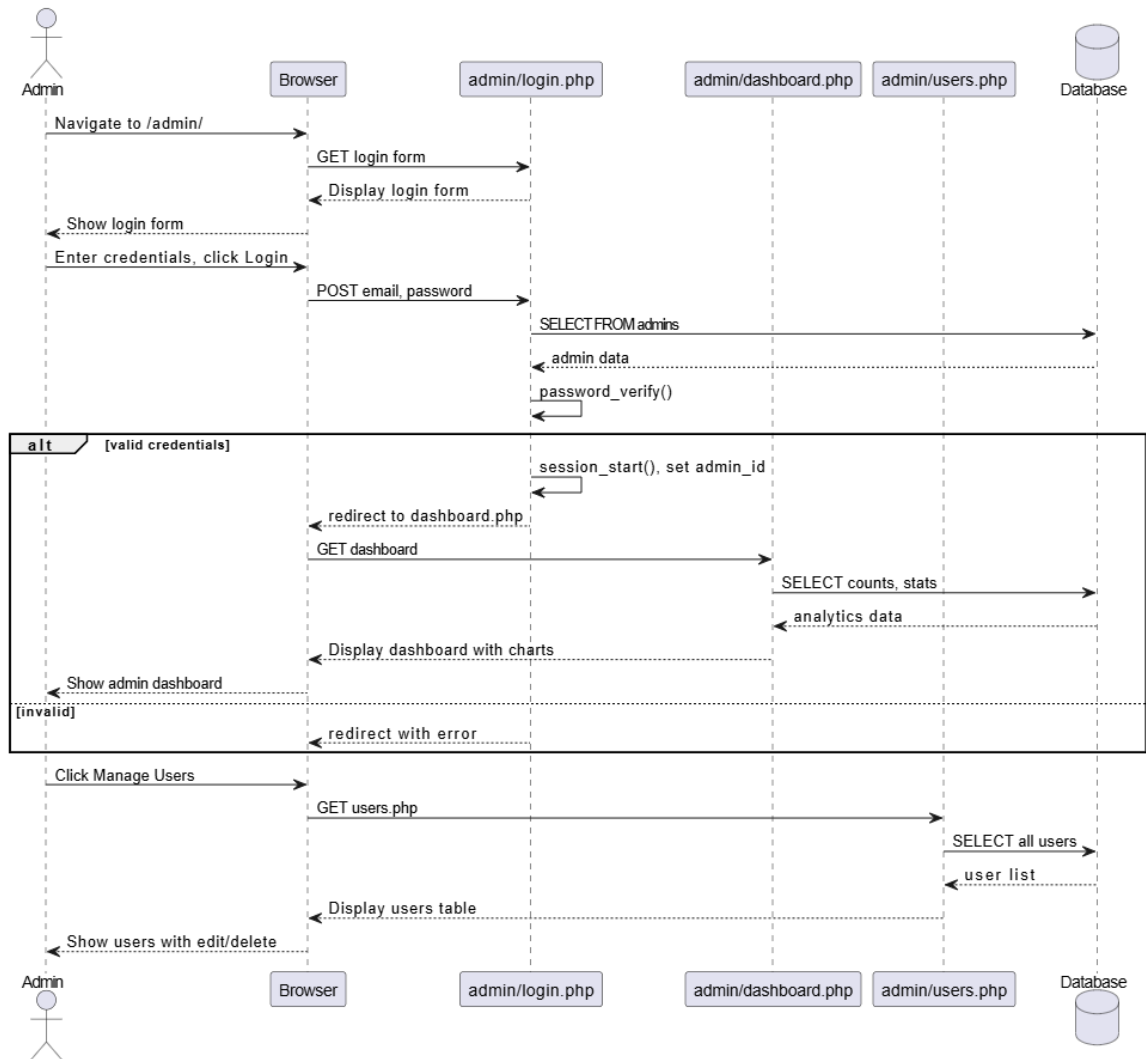


Figure 4-11 Sequence Diagram of Admin Login

4.2.11 Logout

Table 4.11 Logout Use Case

Identifier	UC-11	
Purpose	Logout	
Priority	High	
Pre-condition	User or admin must be logged in to the system.	
Post-condition	User or admin session is terminated successfully.	
Typical Course of Action		
S#	Actor Action	System Response
1	User or admin clicks on the Logout button.	System receives the logout request.
2	System processes the logout action.	System destroys the active session.
3	Logout is completed successfully.	System redirects the user or admin to the login/landing page.
Alternate Course of Action		
S#	Actor Action	System Response
1	If the session is already expired.	System redirects the user or admin to the login/landing page.
2	If user tries to access a protected page after logout.	System denies access and redirects to the login page.

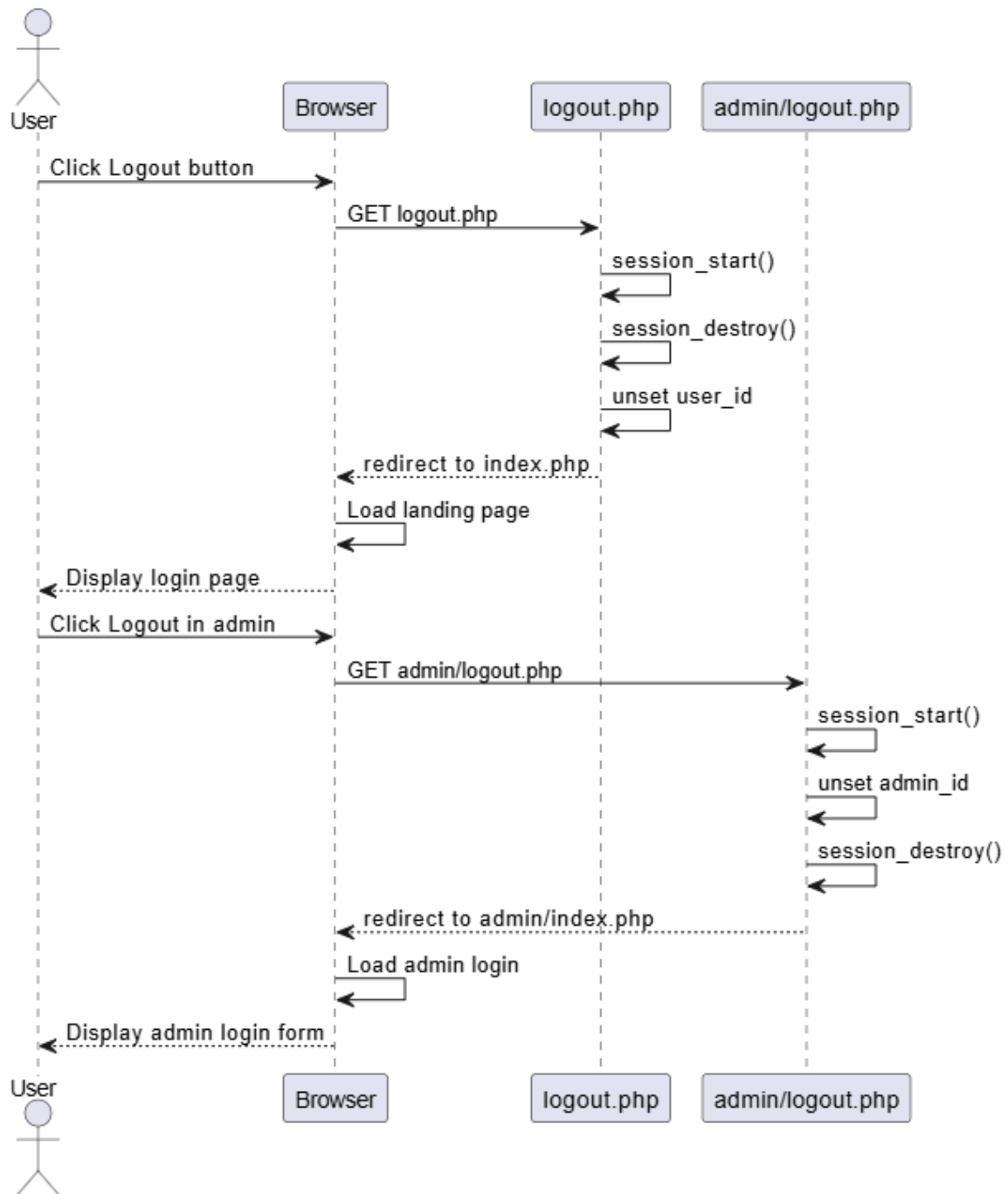


Figure 4-12 Sequence Diagram of Logout

4.3 System Sequence Diagram

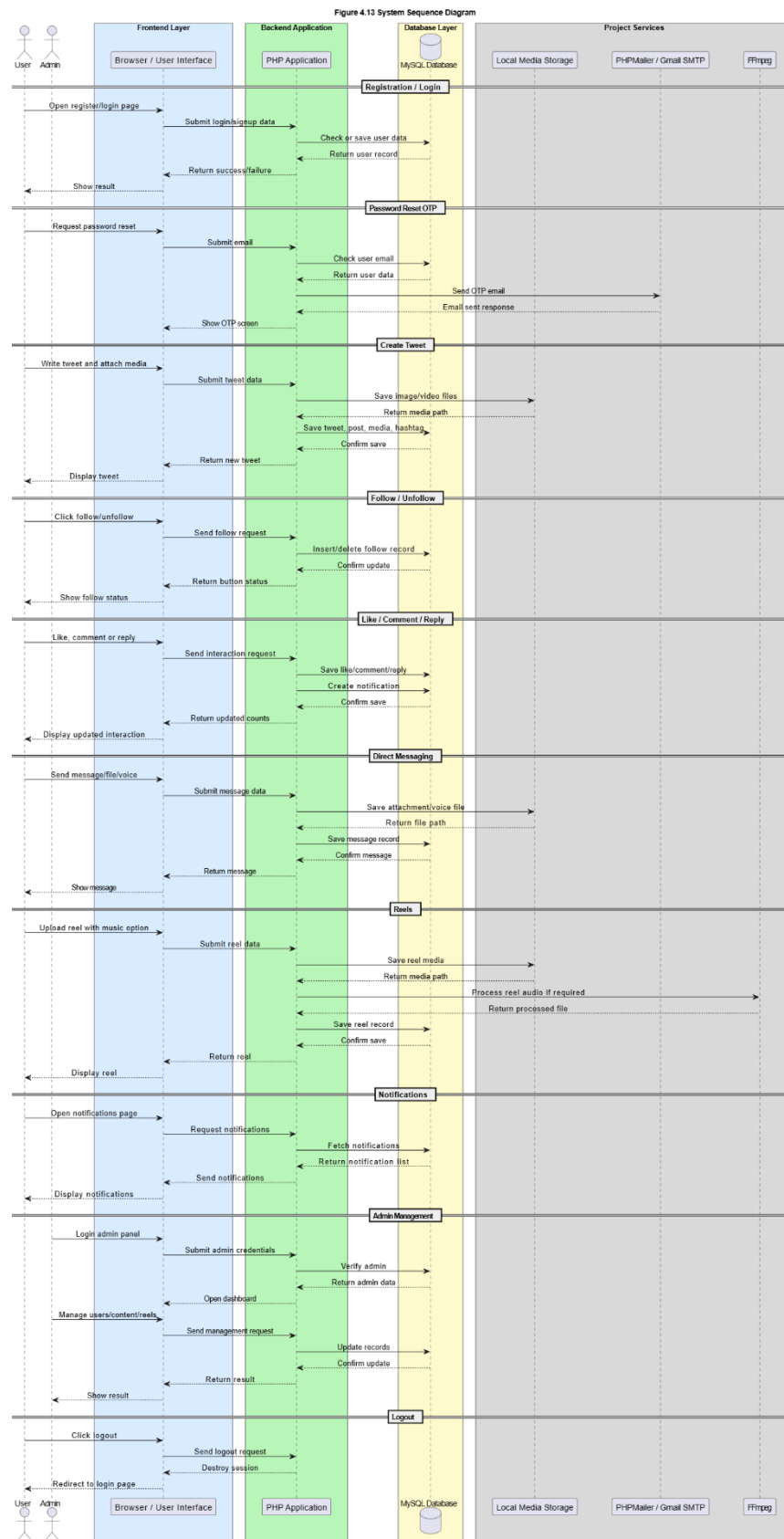


Figure 4-13 System Sequence Diagram

4.4 Class Diagram

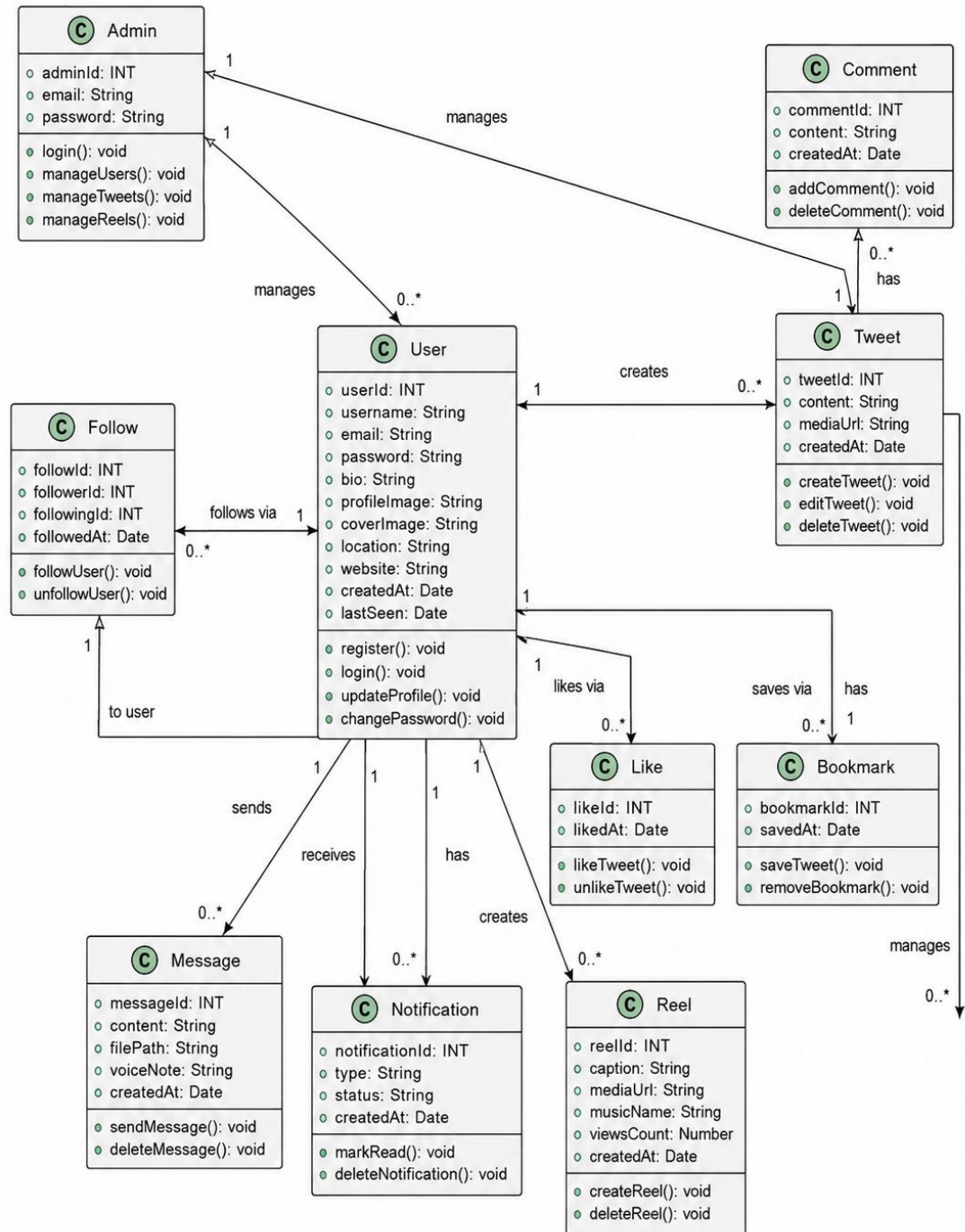


Figure 4-14 Class Diagram

4.5 Data Flow Diagram

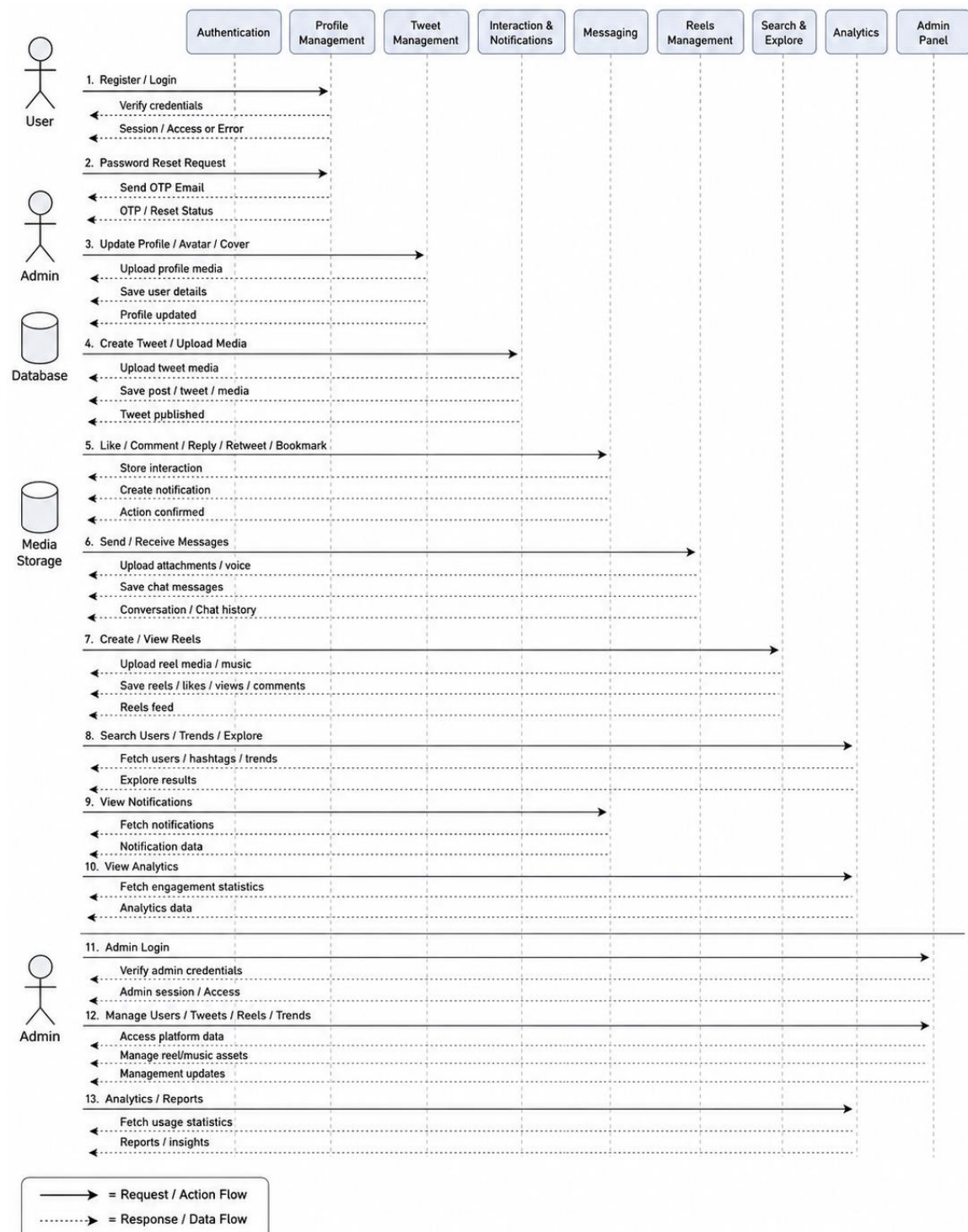


Figure 4-15 Data Flow Diagram of System

CHAPTER NO 5 IMPLEMENTATION AND TESTING

5.1 Testing Methods

In software development, testing is an essential phase used to verify the quality, functionality, security, and reliability of the system. For TwitterClone, different testing methods were applied to ensure that core features such as user authentication, tweeting, following, messaging, reels, notifications, bookmarks, and admin management work correctly.

5.1.1 Black Box Testing

Black box testing focuses on checking the functionality of the system without looking at the internal code structure. In TwitterClone, this testing is used to verify user-facing features.

Objectives:

- Validate user requirements.
- Test login, registration, tweeting, messaging, reels, and admin functions.
- Ensure correct output for valid and invalid inputs.

Example:

- **Input:** Enter a valid email and password on the login page.
- **Expected Output:** User logs in successfully and is redirected to the home feed.

5.1.2 White Box Testing

White box testing examines the internal logic and code structure of the system. It helps verify database queries, PHP functions, validation rules, and session handling.

Objectives:

- Verify code flow and logic.
- Test conditions, loops, and database operations.
- Identify security issues in authentication and user input handling.

Example:

- Test password verification using `password_hash()` and `password_verify()`.
- Validate that prepared statements are used for database operations.

5.1.3 Unit Testing

Unit testing focuses on testing individual modules or functions of the system separately.

Objectives:

- Test authentication functions.
- Test tweet creation logic.
- Test follow/unfollow functionality.
- Test like, comment, bookmark, and notification functions.

Tools Used:

- Manual PHP testing.
- Browser testing.
- Database verification through phpMyAdmin.

5.1.4 Integration Testing

Integration testing checks how different modules work together.

Objectives:

- Test interaction between users, tweets, likes, comments, and notifications.
- Verify that messaging works with user accounts.
- Ensure reels interact correctly with likes, saves, comments, and views.

Example:

- When a user likes a tweet, the like count should update and a notification should be generated for the tweet owner.

5.1.5 System Testing

System testing evaluates the complete application as a whole.

Objectives:

- Test the full workflow of TwitterClone.
- Validate end-to-end features such as signup, login, tweet creation, interaction, messaging, reels, and logout.

Example:

- Verify that a user can register, log in, create a tweet, like another tweet, send a message, view reels, and log out successfully.

5.1.6 Acceptance Testing

Acceptance testing ensures that the system meets user and project requirements.

Objectives:

- Confirm that the system provides a Twitter-like social media experience.
- Ensure all important features are usable and working correctly.
- Verify that admin features support platform management.

Example:

- Users should be able to post tweets, follow users, interact with content, and use messaging without difficulty.

5.1.7 Performance Testing

Performance testing checks the speed, responsiveness, and stability of the system.

Objectives:

- Measure page loading speed.
- Test feed loading, search, reels, and messaging performance.

- Identify slow database queries or heavy pages.

Tools Used:

- Browser Developer Tools.
- Lighthouse.
- Manual load testing.

Example:

- Test how the home feed and reels page behave when many tweets or reels are available.

5.1.8 Security Testing

Security testing identifies vulnerabilities and ensures the system is protected against unauthorized access.

Objectives:

- Verify secure login and password handling.
- Test protection of admin pages.
- Check input validation against SQL injection and invalid uploads.
- Ensure private messages are not accessible by unauthorized users.

Tools Used:

- Manual security testing.
- OWASP ZAP.
- Browser inspection tools.

Example:

- Test login fields and search fields for SQL injection attempts.

5.1.9 Compatibility Testing

Compatibility testing ensures that the system works across different browsers, devices, and screen sizes.

Objectives:

- Test the website on desktop and mobile screens.
- Verify layout responsiveness.
- Check browser compatibility.

Example:

- Verify that home feed, profile page, messages, and reels display correctly on Chrome, Firefox, Edge, and mobile browsers.

5.2 Test Cases

5.2.1 Registration

Table 5.1 Registration Test Case

Field	Description
Identifier	TC-1
Priority	High
Description	To create a new user account.
Pre-conditions	User is not already registered and the system is running.
Post-conditions	User account is created successfully.
Input Data	Name, username, email, password, confirm password.
Detailed Steps	1. Open the signup form. 2. Enter required user details. 3. Click the signup button. 4. System validates the input and creates the account.
Expected Results	1. User is registered successfully. 2. Registration fails if required fields are empty. 3. Registration fails if email or username already exists.
Actual Result	User registered successfully.

5.2.2 Login

Table 5.2 Login Test Case

Field	Description
Identifier	TC-2
Priority	High
Description	To authenticate a registered user.
Pre-conditions	User must already be registered.
Post-conditions	User logs in and is redirected to the home page.
Input Data	Email and password.
Detailed Steps	1. Open the login page. 2. Enter registered email and password. 3. Click the login button.4. System verifies the credentials.
Expected Results	1. User logs in successfully if credentials are correct. 2. Login fails if email or password is incorrect. 3. Login fails if required fields are empty.
Actual Result	User logged in successfully.

5.2.3 Create Tweet

Table 5.3 Create Tweet Test Case

Field	Description
Identifier	TC-3
Priority	High
Description	To allow users to create a tweet.
Pre-conditions	User must be logged in.
Post-conditions	Tweet is posted and displayed in the feed.
Input Data	Tweet text, image, or video.
Detailed Steps	1. Navigate to the home page.2. Enter tweet content.3. Optionally upload image or video.4. Click the tweet/post button.
Expected Results	1. Tweet is posted successfully.2. Empty tweet is not submitted.3. Invalid media files are rejected.
Actual Result	Tweet created successfully.

5.2.4 Update Profile

Table 5.4 Update Profile Test Case

Field	Description
Identifier	TC-4
Priority	High
Description	To allow users to update profile information.
Pre-conditions	User must be logged in.
Post-conditions	Profile details are updated successfully.
Input Data	Name, bio, location, website, profile image, cover image.
Detailed Steps	1. Open the profile page. 2. Click edit profile. 3. Update profile information. 4. Upload profile or cover image if required. 5. Save changes.
Expected Results	1. Profile is updated successfully. 2. Invalid data is rejected. 3. Uploaded image is displayed correctly.
Actual Result	Profile updated successfully.

5.2.5 Like Tweet

Table 5.5 Like Tweet Test Case

Field	Description
Identifier	TC-5
Priority	Medium
Description	To allow users to like or unlike a tweet.
Pre-conditions	User must be logged in and tweet must exist.
Post-conditions	Like status and like count are updated.
Input Data	Click on like button.
Detailed Steps	1. Open home feed or profile feed. 2. Click the like button on a tweet. 3. Click again to unlike.
Expected Results	1. Tweet is liked successfully. 2. Like count increases. 3. Unlike decreases the like count. 4. Notification is generated for the tweet owner.
Actual Result	Like functionality works successfully.

5.2.6 Comment and Reply

Table 5.6 Comment and Reply Test Case

Field	Description
Identifier	TC-6
Priority	High
Description	To allow users to comment and reply on tweets.
Pre-conditions	User must be logged in and tweet must exist.
Post-conditions	Comment or reply is saved and displayed.
Input Data	Comment text or reply text.
Detailed Steps	1. Open a tweet or status page. 2. Enter a comment. 3. Submit the comment. 4. Reply to an existing comment if needed.
Expected Results	1. Comment is added successfully. 2. Reply is added successfully. 3. Empty comments are not submitted. 4. Notification is generated for the related user.
Actual Result	Comment and reply added successfully.

5.2.7 Follow and Unfollow User

Table 5.7 Follow and Unfollow Test Case

Field	Description
Identifier	TC-7
Priority	High
Description	To allow users to follow and unfollow other users.
Pre-conditions	User must be logged in and another user profile must exist.
Post-conditions	Follow status is updated.
Input Data	Follow or unfollow button click.
Detailed Steps	1. Open another user's profile. 2. Click the follow button. 3. Click the unfollow button if already following.
Expected Results	1. User follows another user successfully. 2. Follower and following counts update. 3. Unfollow removes the relation. 4. Notification is generated on follow.
Actual Result	Follow/unfollow works successfully.

5.2.8 5.2.8 Bookmark Tweet

Table 5.8 Bookmark Test Case

Field	Description
Identifier	TC-8
Priority	Medium
Description	To allow users to save tweets for later.
Pre-conditions	User must be logged in and tweet must exist.
Post-conditions	Tweet is saved in bookmarks.
Input Data	Bookmark button click.
Detailed Steps	1. Open the feed. 2. Click bookmark on a tweet. 3. Open bookmarks page. 4. Verify the saved tweet.
Expected Results	1. Tweet is bookmarked successfully. 2. Bookmarked tweet appears on bookmarks page. 3. Clicking again removes bookmark.
Actual Result	Bookmark functionality works successfully.

5.2.9 Direct Messaging

Table 5.9 Direct Messaging Test Case

Field	Description
Identifier	TC-9
Priority	High
Description	To allow users to send private messages.
Pre-conditions	Sender and receiver must be registered users.
Post-conditions	Message is delivered and displayed in chat.
Input Data	Text message, attachment, or voice note.
Detailed Steps	1. Open messages page. 2. Select a user conversation. 3. Type a message or attach a file. 4. Click send. 5. Verify message appears in conversation.
Expected Results	1. Message is sent successfully. 2. Attachments are uploaded correctly. 3. Voice notes are sent successfully. 4. Blocked users cannot send messages.
Actual Result	Message sent successfully.

5.2.10 Reels

Table 5.10 Reels Test Case

Field	Description
Identifier	TC-10
Priority	High
Description	To allow users to create and interact with reels.
Pre-conditions	User must be logged in.
Post-conditions	Reel is created or interaction is saved.
Input Data	Reel image/video, caption, music, like, save, comment.
Detailed Steps	1. Open create reel page. 2. Upload reel media. 3. Add caption or music if required. 4. Submit reel. 5. Open reels page and interact with reels.
Expected Results	1. Reel is created successfully. 2. User can like, save, comment, and view reels. 3. Invalid media files are rejected.
Actual Result	Reels functionality works successfully.

5.2.11 Notifications

Table 5.11 Notifications Test Case

Field	Description
Identifier	TC-11
Priority	Medium
Description	To verify that users receive notifications for interactions.
Pre-conditions	User must be logged in and another user must interact with their content.
Post-conditions	Notification is displayed in the notification section.
Input Data	Like, follow, comment, reply, retweet, quote, or mention action.
Detailed Steps	1. User A interacts with User B's content. 2. User B opens the notification page. 3. System displays the notification.
Expected Results	1. Notification is generated correctly. 2. Notification can be viewed. 3. User can delete individual or all notifications.
Actual Result	Notifications displayed successfully.

5.2.12 Search and Explore

Table 5.12 Search and Explore Test Case

Field	Description
Identifier	TC-12
Priority	Medium
Description	To verify user search, explore feeds, and trending hashtags.
Pre-conditions	User must be logged in.
Post-conditions	Search or explore results are displayed.
Input Data	Search keyword, hashtag, or explore tab selection.
Detailed Steps	1. Open search or explore page. 2. Enter a username or keyword. 3. Open explore tabs or trending hashtags. 4. Verify displayed results.
Expected Results	1. Relevant users are displayed. 2. Trending hashtags are shown. 3. Explore tabs load correct content.
Actual Result	Search and explore work successfully.

5.2.13 Admin Management

Table 5.13 Admin Management Test Case

Field	Description
Identifier	TC-13
Priority	High
Description	To verify admin dashboard and moderation features.
Pre-conditions	Admin must be logged in with valid credentials.
Post-conditions	Admin actions are applied successfully.
Input Data	User data, tweet data, comment data, reel data, trend data.
Detailed Steps	1. Open admin login page. 2. Enter admin credentials. 3. Access admin dashboard. 4. Manage users, tweets, comments, replies, quotes, reels, music, bookmarks, and trends.
Expected Results	1. Admin logs in successfully. 2. Admin can view dashboard statistics. 3. Admin can manage users and content. 4. Unauthorized users cannot access admin pages.
Actual Result	Admin management works successfully.

5.2.14 Logout

Table 5.14 Logout Test Case

Field	Description
Identifier	TC-14
Priority	High
Description	To allow users and admins to logout from the system.
Pre-conditions	User or admin must be logged in.
Post-conditions	Session is terminated and user is redirected.
Input Data	Logout button click.
Detailed Steps	1. Click the logout button. 2. System clears the session. 3. User is redirected to login or landing page. 4. Try accessing protected pages again.
Expected Results	1. User logs out successfully. 2. Session is destroyed. 3. Protected pages cannot be accessed without login.
Actual Result	Logout works successfully.

CHAPTER NO 6 USER MANUAL

6.1 Landing and Login Page

The Landing and Login page is the first screen of the TwitterClone web application. It presents the platform introduction on the left side and the login form on the right side. Registered users can enter their email address and password to access their accounts, while new users can use the Sign Up option to create an account. The page also provides a Forgot Password option for account recovery. This screen acts as the main entry point for both existing and new users..

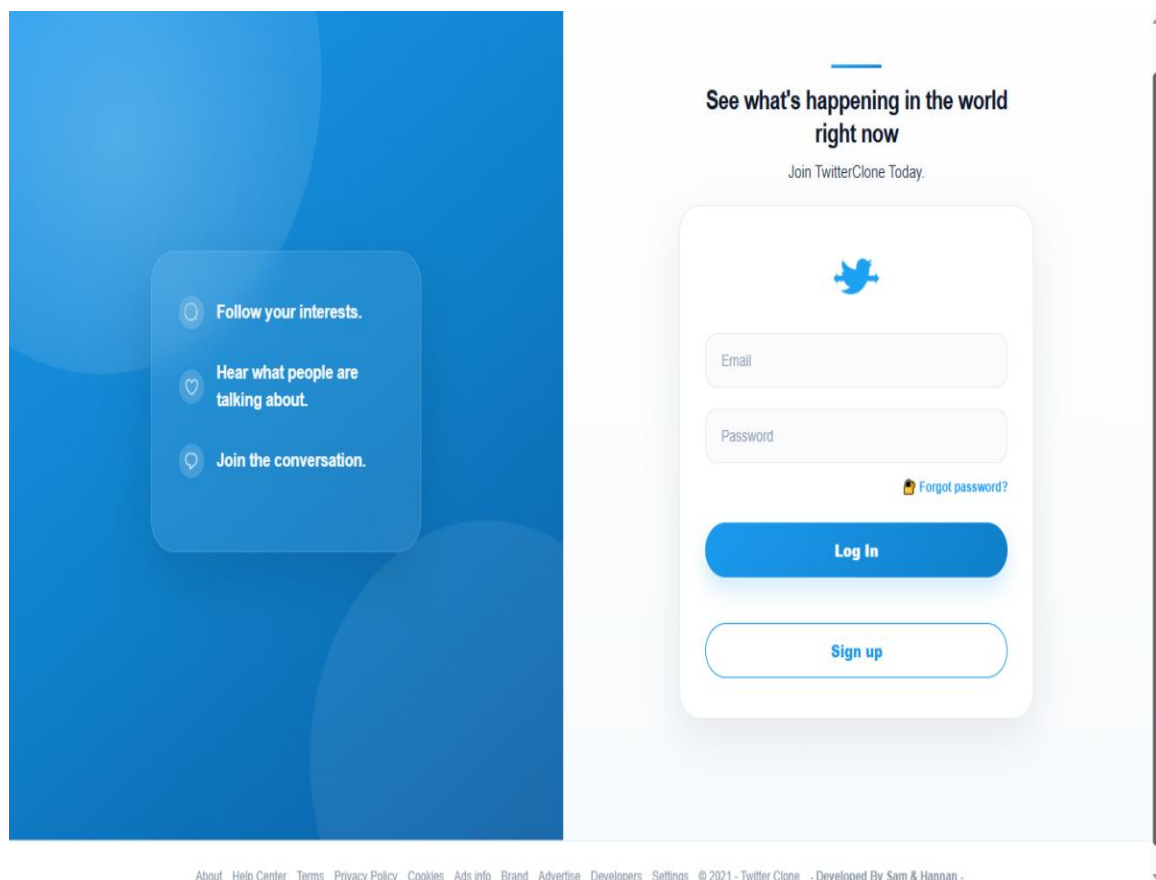
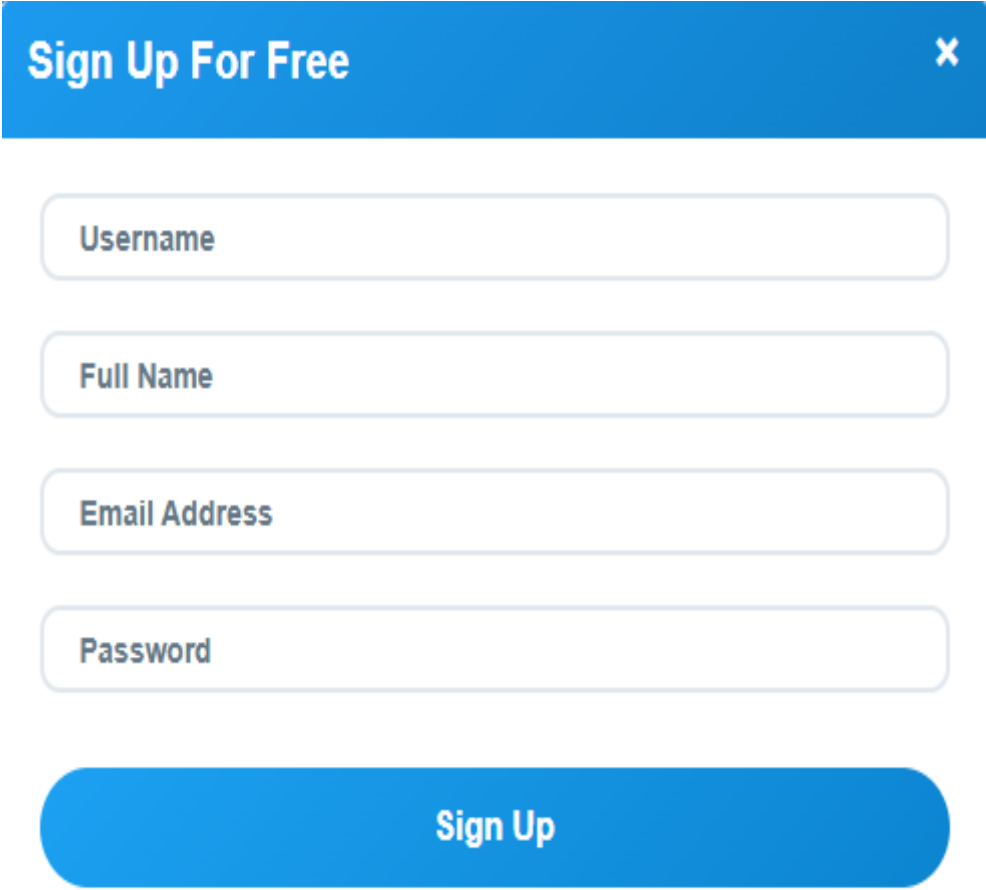


Figure 6-1 Landing Page of TwitterClone

6.2 Sign Up Page

The Sign Up page allows new users to create an account on TwitterClone. To register, the user enters a unique username, full name, email address, and password in the provided fields. After clicking the Sign Up button, a 6-digit verification code is sent to the registered email address. The user must enter the verification code within two minutes to complete the registration process. Once the code is successfully verified, the account is created and the user can access the platform. If the selected username or email address already exists, the system displays an appropriate error message and prompts the user to enter different information..



The image shows a user registration form with a blue header bar containing the text "Sign Up For Free" and a close button (X). Below the header are four input fields: "Username", "Full Name", "Email Address", and "Password". At the bottom is a large blue button labeled "Sign Up".

Sign Up For Free X

Username

Full Name

Email Address

Password

Sign Up

Figure 6-2 User Registration Page

6.3 Dashboard Page

The Home Page serves as the main dashboard of TwitterClone after a user logs into the platform. It provides a personalized feed where users can view tweets from accounts they follow, create new tweets with text and media, and interact with posts through likes, comments, retweets, and bookmarks. The left sidebar offers quick navigation to major modules such as Explore, Reels, Messages, Notifications, Bookmarks, Profile, and Settings. The right sidebar displays user suggestions and trending hashtags, helping users discover new content and connect with other members of the platform. This page acts as the central hub for social interaction and content sharing within TwitterClone.

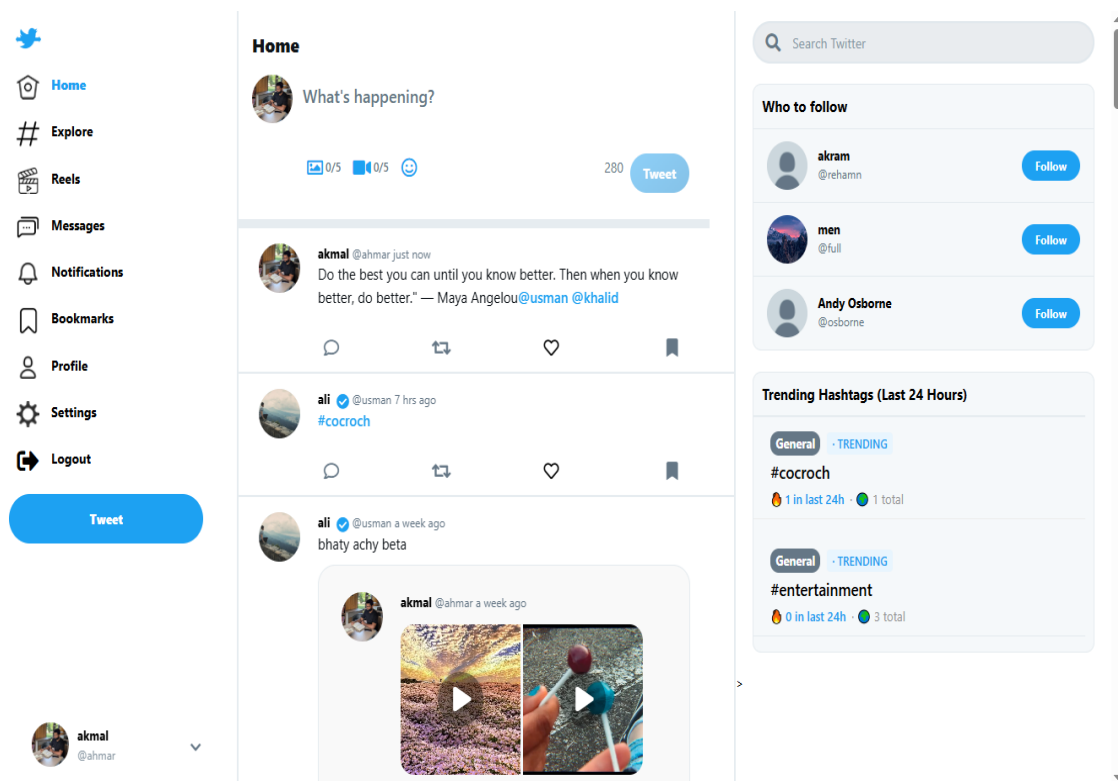


Figure 6-3Home Page Dashboard

6.4 Profile Page

The Profile Page allows users to view and manage their personal information on the TwitterClone platform. It displays the user's profile picture, cover photo, name, username, bio, join date, location, website, follower count, and following count. Users can update their profile details through the Edit Profile option and customize their account appearance. The page also organizes user activity into separate sections, including Tweets, Tweets & Replies, Media, and Likes, making it easier to browse and manage shared content. This page serves as the user's personal identity and activity hub within the platform.



Figure 6-4 User Profile Page

6.5 Explore Page

The Explore Page helps users discover new content, trending topics, and other members of the TwitterClone platform. It provides a search feature for finding users and posts, along with multiple content categories such as For You, Latest, Top, News, People, Photos, and Videos. Users can browse personalized recommendations, view media-rich content, and explore trending discussions based on their interests. This page enhances content discovery and helps users stay engaged with the latest activity across the platform.

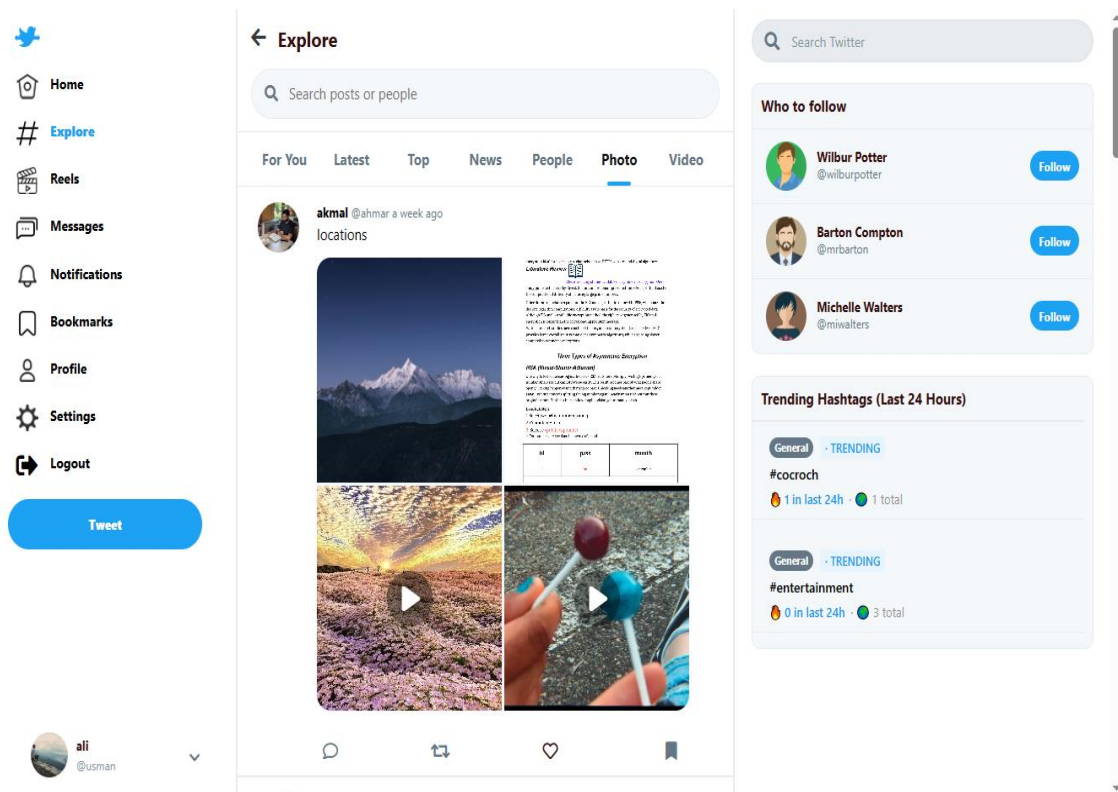


Figure 6-5 Explore Page

6.6 Reels Page

The Reels Page allows users to create, discover, and interact with short-form video content on the TwitterClone platform. Users can browse reels through the For You and Following feeds, search reels using hashtags, and view personalized content recommendations. The page supports engagement features such as likes, comments, saves, shares, views, and follow actions. Users can also create and upload their own reels with captions and optional music, making the platform more interactive and engaging through visual content.

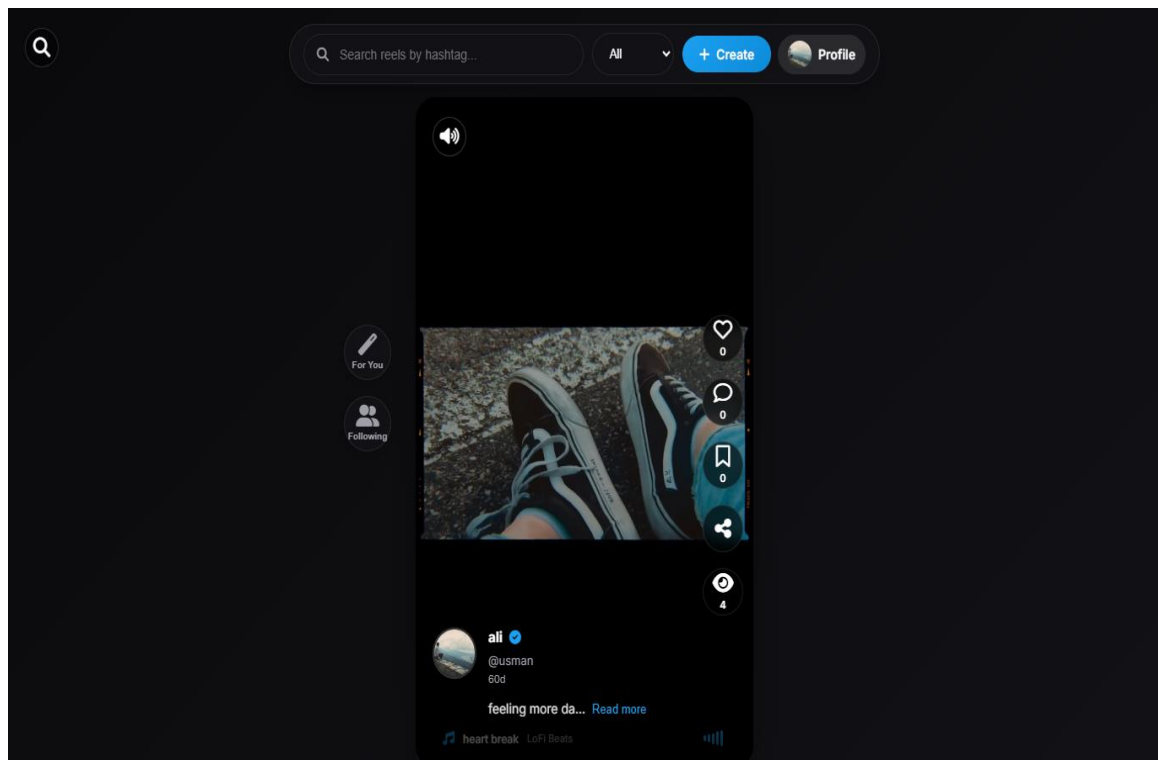


Figure 6-6Reels Page

6.7 Messages Page

The Messages Page allows users to have private conversations with other users.

Key Features:

- **Recent Chats Sidebar** – Shows all active conversations with other users. Click any chat to open it.
- **Chat Area** – Displays the full conversation history including text, images, videos, and voice notes with timestamps.
- **User Info Header** – Shows the other user's name and their last seen time.
- **Message Input** – Type text, attach media files, or record voice messages at the bottom, then click Send.
- **Message Actions** – Users can delete messages (for themselves or everyone), reply to specific messages, or pin important messages and download for later use.

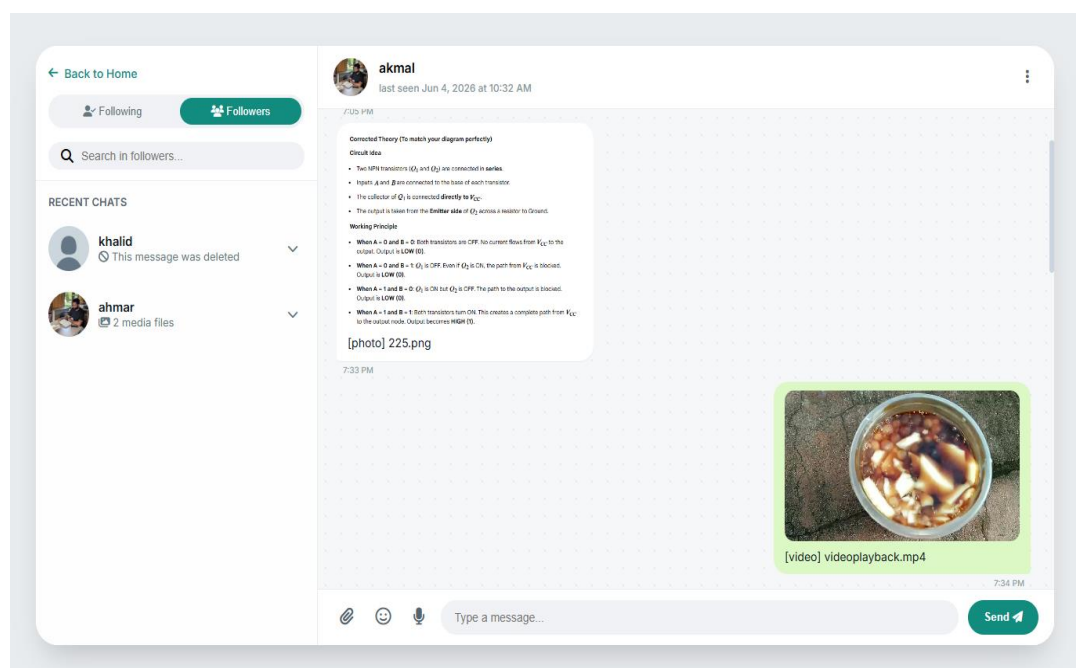


Figure 6-7 Direct Messaging Page

6.8 Bookmarks Page

The Bookmarks Page allows users to save and organize tweets for future reference. It displays all bookmarked posts in a dedicated section, enabling users to quickly access important content without searching through their timeline. Users can view text, images, and videos from saved tweets and continue interacting with them through likes, comments, and shares. This feature helps users manage and revisit valuable content whenever needed.

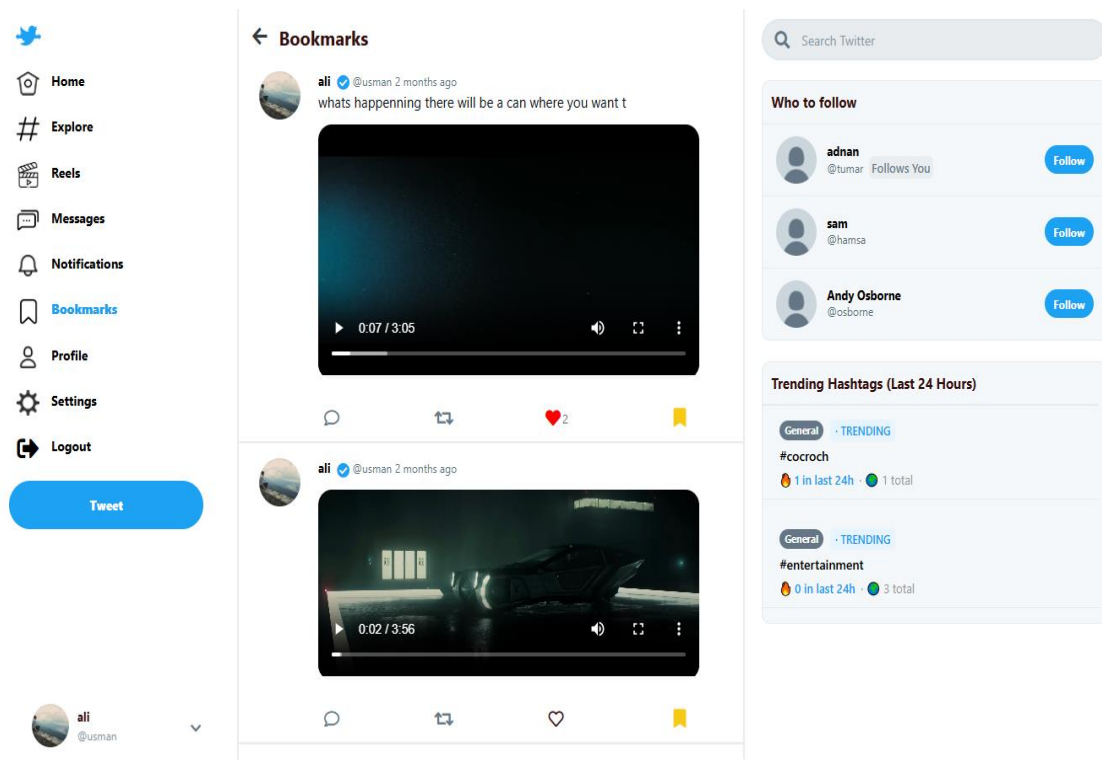


Figure 6-8Bookmarks Page

6.9 Notifications Page

The Notifications Page keeps users informed about activities related to their accounts and content on the TwitterClone platform. It displays real-time notifications for important events such as likes, comments, replies, mentions, follows, retweets, and other user interactions. Users can review recent activity, access related content directly from notifications, and remove individual notifications or clear all notifications when needed. This page helps users stay updated and engaged with the latest interactions occurring across the platform.

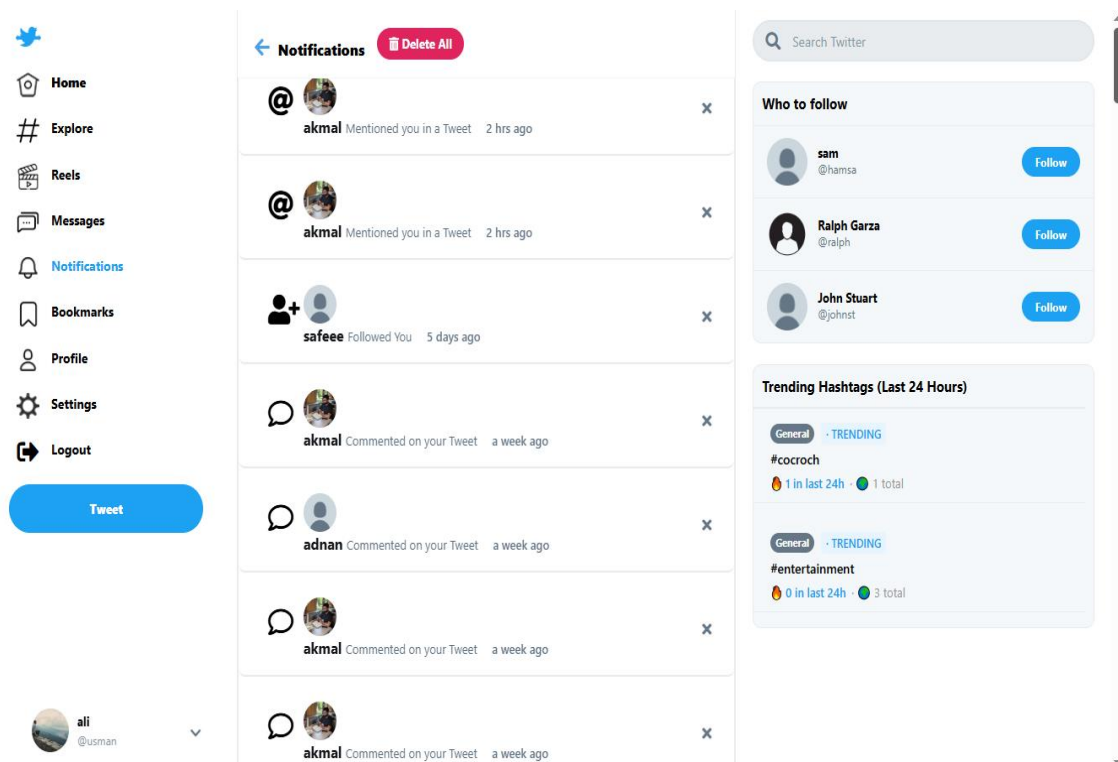


Figure 6-9 Notifications Page

6.10 Status Page

The Status Page displays the complete details of a selected tweet and provides a dedicated view for user interactions. Users can view the tweet content, author information, posting date and time, and engagement options such as likes, comments, retweets, bookmarks, and sharing. The page also allows users to participate in discussions by viewing and responding to comments and replies related to the tweet. This page serves as the central location for managing and interacting with individual posts on the TwitterClone platform.

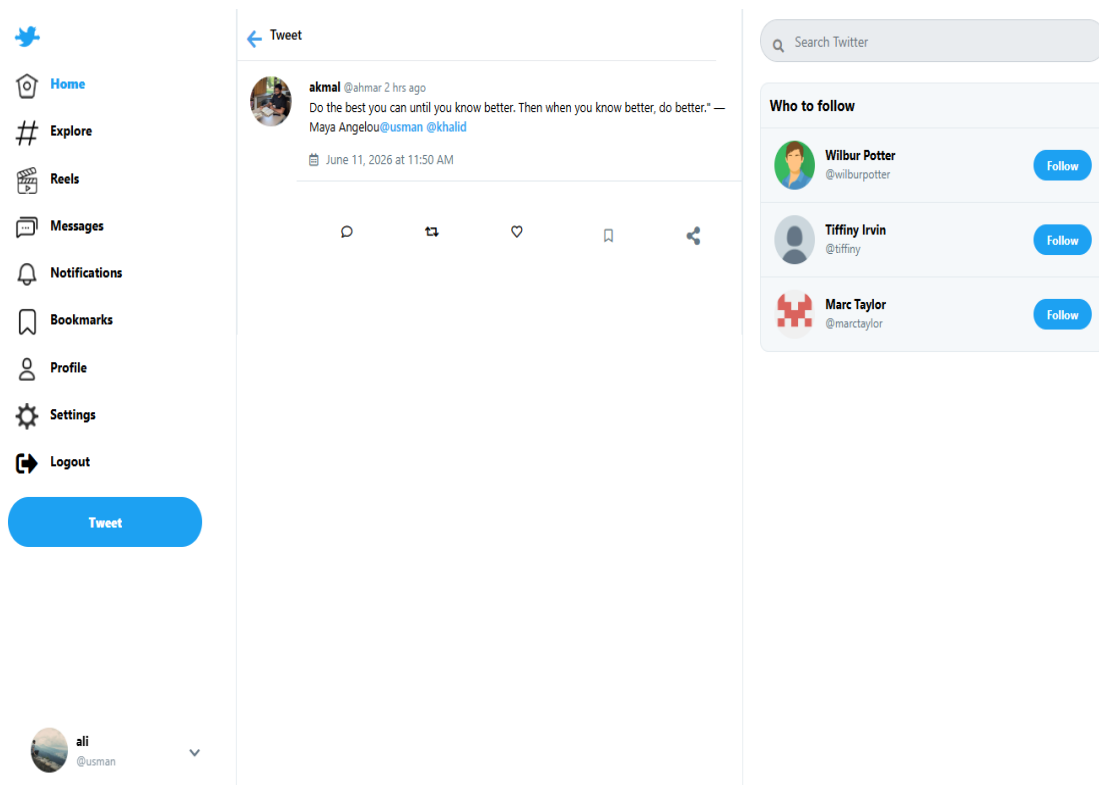


Figure 6-10 Tweet Status Page

6.11 Account Settings

The Account Settings Page allows users to manage and update important account information on the TwitterClone platform. Users can change their username and password through dedicated settings options while maintaining account security and profile integrity. The system validates all entered information before applying changes and provides feedback for successful updates or validation errors. This page helps users personalize and secure their accounts according to their preferences.

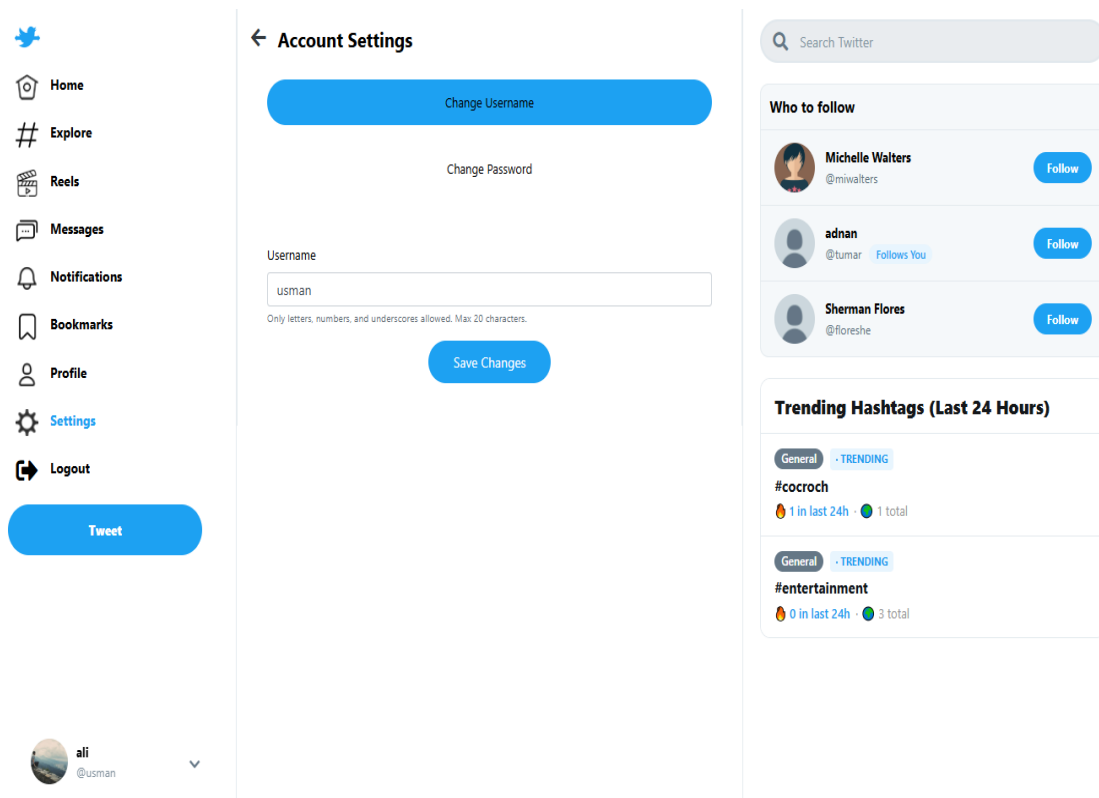


Figure 6-11 Account Settings Page